Rising Market Power: Debates and Case Studies

September, 2026

Where We Are

- In *Firm Conduct* we studied the NEIO program: price-cost margins are **not** observable, so marginal cost must be *inferred* from firm behavior, and conduct must be *identified* from exogenous rotations/shifts of residual demand.
- In *Demand and Welfare* we built the demand-side machinery: substitution patterns, elasticities, and the mapping from estimated demand to consumer surplus.
- This lecture puts the two together and asks an empirical question that has escaped the field: has market power in the US economy risen since 1980?
  - Two very different research strategies have been brought to bear.
  - They disagree. Understanding *why* they disagree is the point of the lecture.

Introduction

- Recall one of the leading "hypotheses" for the declining labor share, consistent with rising concentration, is **increasing markups**.
- Is this a promising avenue to explore? De Loecker, Eeckhout, and Unger (2020, henceforth DEU) argue it is.
  - Headline number: aggregate markup rises from $\mu = 1.21$ to $\mu = 1.61$.
  - Mainly driven by within-industry *reallocation* from low-markup to high-markup firms.
- Subsequent work tends to revise this number down substantially, although some upward trend possibly survives. Four distinct lines of attack:
  - Technological change or markups? (Raval 2022; Demirer 2025)
  - The non-identification of markups from revenue data. (Bond et al. 2021)
  - Overhead cost vs. variable cost. (Traina 2018)
  - What *is* an aggregate markup, and why do we care? (Edmond, Midrigan, Xu 2023)
- And a fifth response: **go get an industry** and do IO properly.

Road-map from here

- **Part I — The production approach and its discontents.** One estimating equation, $\hat{\mu}_{it} = \hat{\theta}^j_{it}/s^j_{it}$, and four things that can go wrong with it.
- **Part II — Why we care.** Aggregation: what object is "the" aggregate markup, and how does its *level* vs. its *dispersion* map into allocative efficiency?
- **Part III — Case study 1: automobiles** (Grieco, Murry, and Yurukoglu 2024). Full demand-side IO over 1980–2018. Markups *fall*; consumer surplus rises. Directly confronts DEU on the same industry.
- **Part IV — Case study 2: beer** (Miller and Weinberg 2017). Markups are not just a demand-elasticity object: *conduct* moves too, and mergers move it.
- **Part V — Synthesis.** What each lens can and cannot answer.

De Loecker, Eeckhout, and Unger, 2020

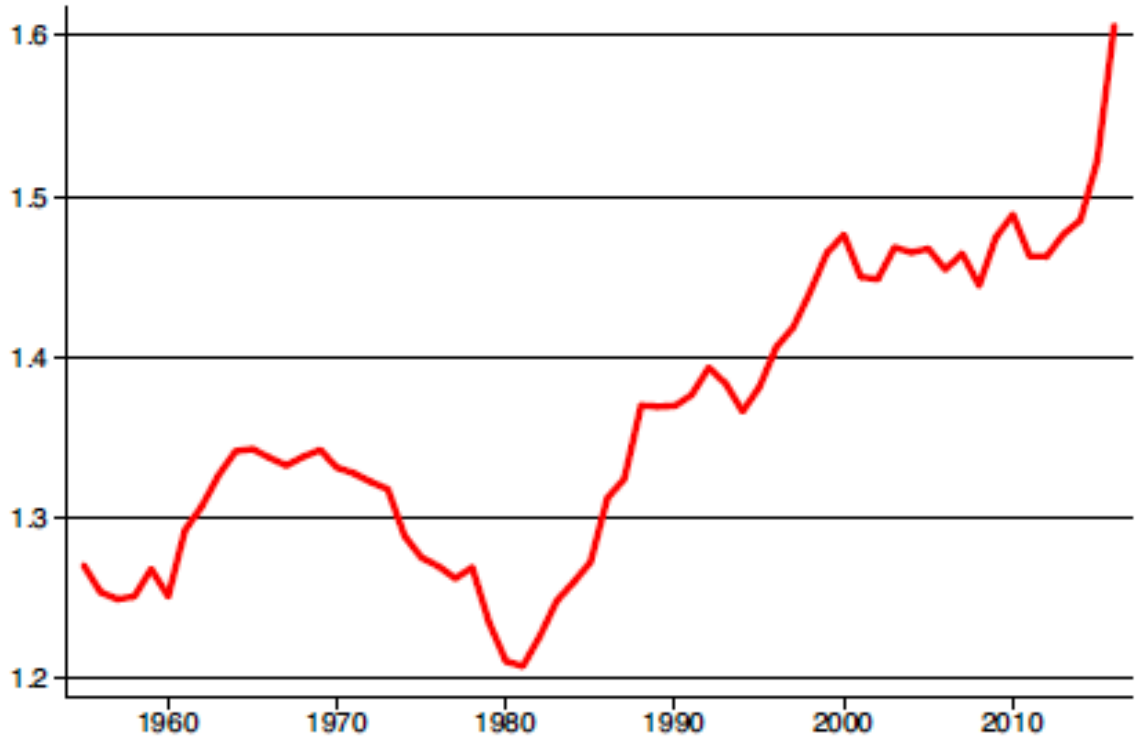
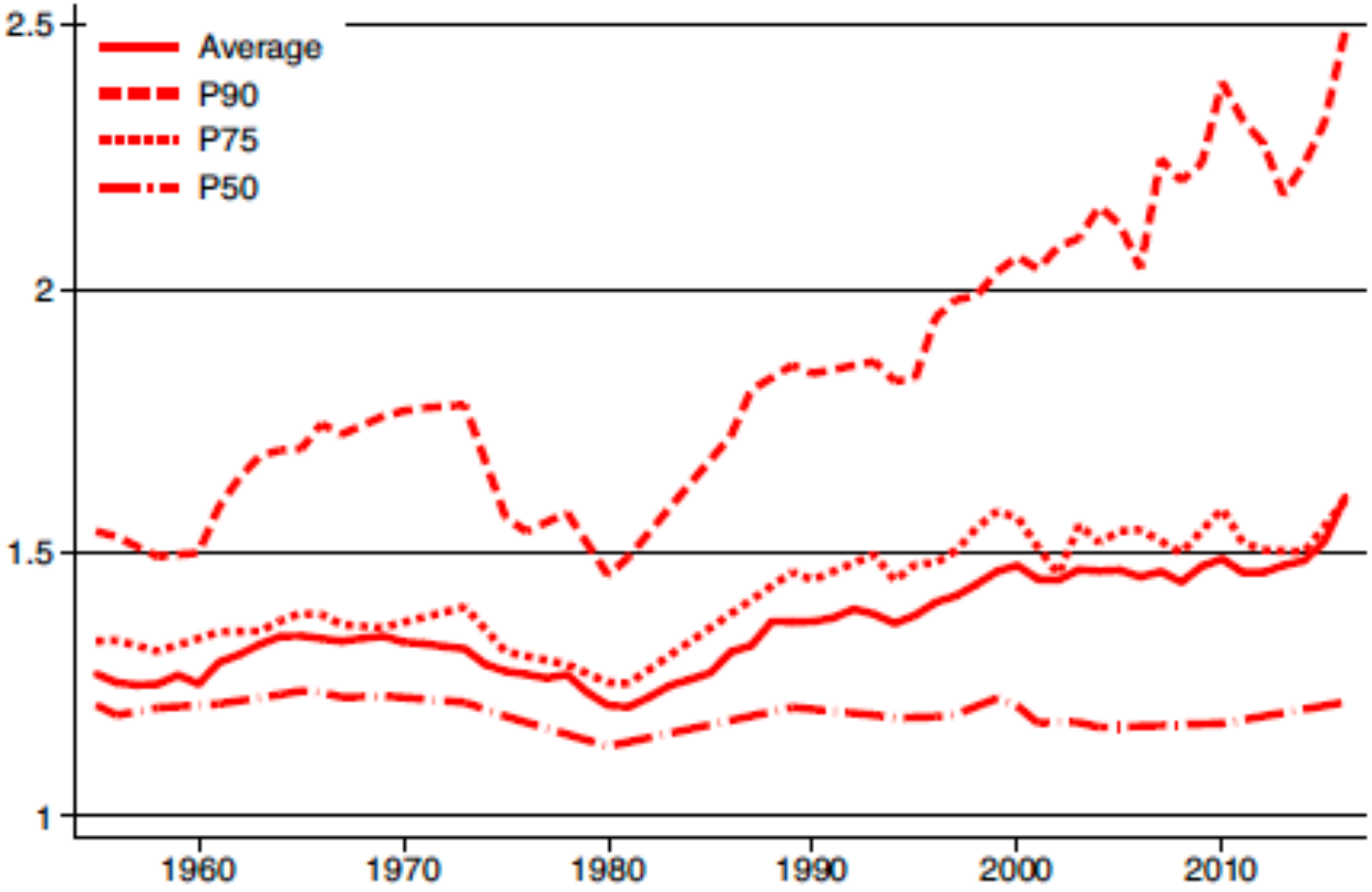


FIGURE I

Average Markups

Output elasticities θ_{st} from the estimated production function are time-varying and sector-specific (two-digit). The average is revenue weighted. The figure illustrates the evolution of the average markup from 1955 to 2016.

De Loecker, Eeckhout, and Unger, 2020



(B) Percentiles markup distribution (revenue weight)

Measurement Framework: Production Approach

- Pioneered by Hall (1988) and adapted by De Loecker and Warzynski (2012) to firm-level data.
- Consider a producer with $Q_{it} = F_{it}(K_{it}, X_{it}^1, .., X_{it}^J)$; cost minimization implies

$$\lambda_{it} \frac{\partial F_{it}}{\partial X_{it}^j} = W_{it}^j, \quad \forall j = 1, .., J$$

  λ_{it} is the marginal cost (i.e. the Lagrange multiplier on output)
- Convert to elasticity form, we obtain

$$\theta_{it}^j \equiv \frac{\partial F_{it}}{\partial X_{it}^j} \frac{X_{it}^j}{Q_{it}} = \frac{P_{it}}{\lambda_{it}} \left(\frac{W_{it}^j X_{it}^j}{P_{it} Q_{it}} \right) \equiv \mu_{it} \, s_{it}^j$$

- If one could consistently estimate the output elasticity θ_{it}^j and observe the input revenue share s_{it}^j, we can uncover the markup as

$$\hat{\mu}_{it} = \frac{\hat{\theta}_{it}^j}{s_{it}^j}$$

- Note what is attractive: no demand system, no conduct assumption, no market definition. That is exactly why it scales to the whole economy — and exactly where the vulnerabilities are.

What the Production Approach Has to Get Right

The estimator $\hat{\mu}_{it} = \hat{\theta}^j_{it} / s^j_{it}$ is only as good as four maintained assumptions. Each one organizes a branch of the critique literature:

1. X^j **is truly flexible, and is the *only* margin.** If technology is input-augmenting, different j deliver different $\hat{\mu}$. $\rightarrow$ *Raval (2022), Demirer (2025)*
2. θ^j **is an *output* elasticity.** With revenue data we estimate a *revenue* elasticity, which is contaminated by the very markup we want. $\rightarrow$ *Bond et al. (2021)*
3. s^j **is the *variable* cost share.** Accounting categories (COGS vs. SG&A) do not respect the fixed/variable distinction. $\rightarrow$ *Traina (2018)*
4. **Firm markups aggregate in an economically meaningful way.** $\rightarrow$ *Edmond, Midrigan, and Xu (2023)*

Keep a scorecard: for each critique, ask whether it moves the *level* of μ, the *trend* in μ, or both. We will fill it in at the end of Part I.

Raval 2022: The Same Firm, Two Different Markups

- In principle the procedure works for *any* variable input. Raval (2022) runs it separately on labor and materials and finds the two markups differ substantially.

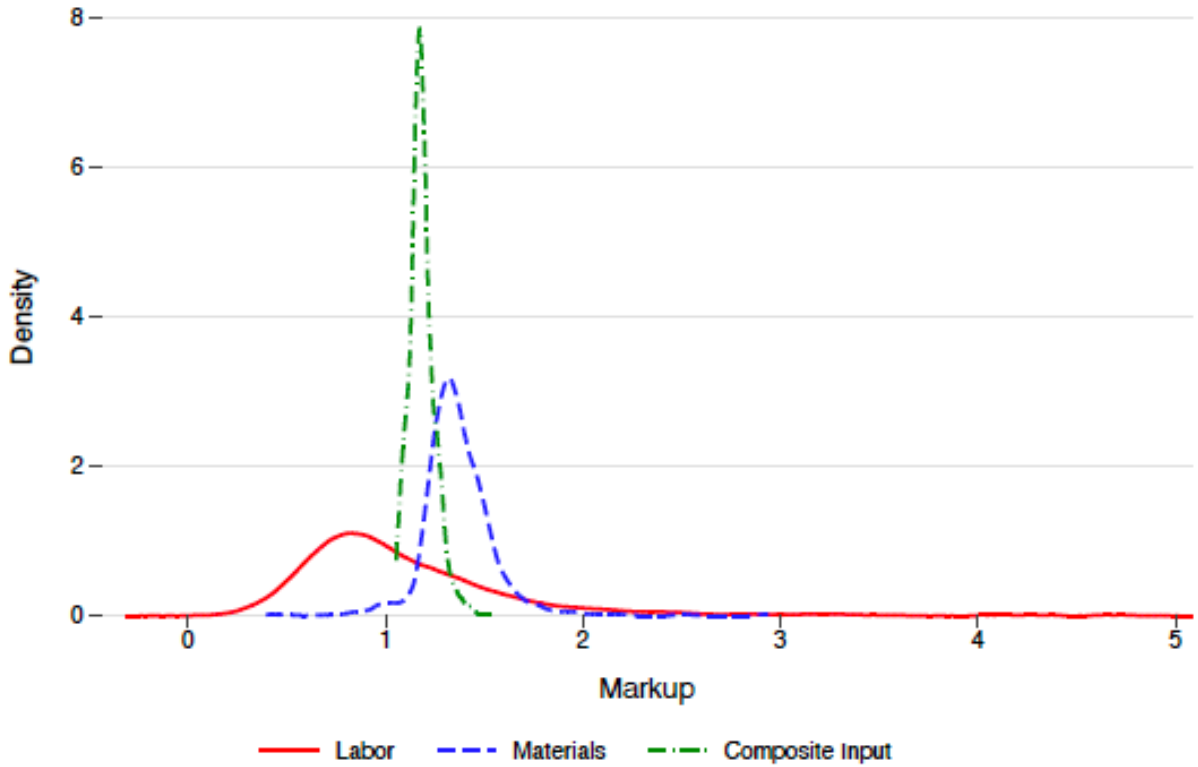


Figure 1 Distribution of Translog Markups for Chilean Food Products, 1996

Raval 2022: They Are Not Just Noisy, They Move Opposite

- Broadly speaking labor markup and material markup are *negatively* correlated — not what classical measurement error would deliver.

Table III Relationship between Markup Estimates

Dataset	Cobb-Douglas	Translog
Chile	-0.66	-0.16
	(0.017)	(0.014)
Colombia	-0.99	-0.28
	(0.015)	(0.021)
India	-1.73	-0.53
	(0.012)	(0.009)
Indonesia	-0.97	-0.48
	(0.018)	(0.021)
US	-0.20	-0.50
	(0.032)	(0.029)
S Europe	-0.80	-0.32
	(0.005)	(0.008)
Retailer	-7.51	-10.08
	(0.143)	(0.102)

Note: Estimates based on (6) where the labor markup is the dependent variable and materials markup the independent variable. Standard errors are clustered at the establishment level.

Raval 2022: Factor-Augmenting Productivity

- The puzzle is that the output-elasticity ratio is not aligned with the factor-share ratio

$$\frac{\hat{\mu}_{it}^{L}}{\hat{\mu}_{it}^{M}} = \frac{\hat{\theta}_{it}^{L}}{\hat{\theta}_{it}^{M}} \frac{s_{it}^{M}}{s_{it}^{L}} \neq 1$$

- Raval (2022) proposes a simple explanation: **factor-augmenting productivity**

$$Q_{it} = F_{it}(K_{it}, B_{it}L_{it}, M_{it})$$

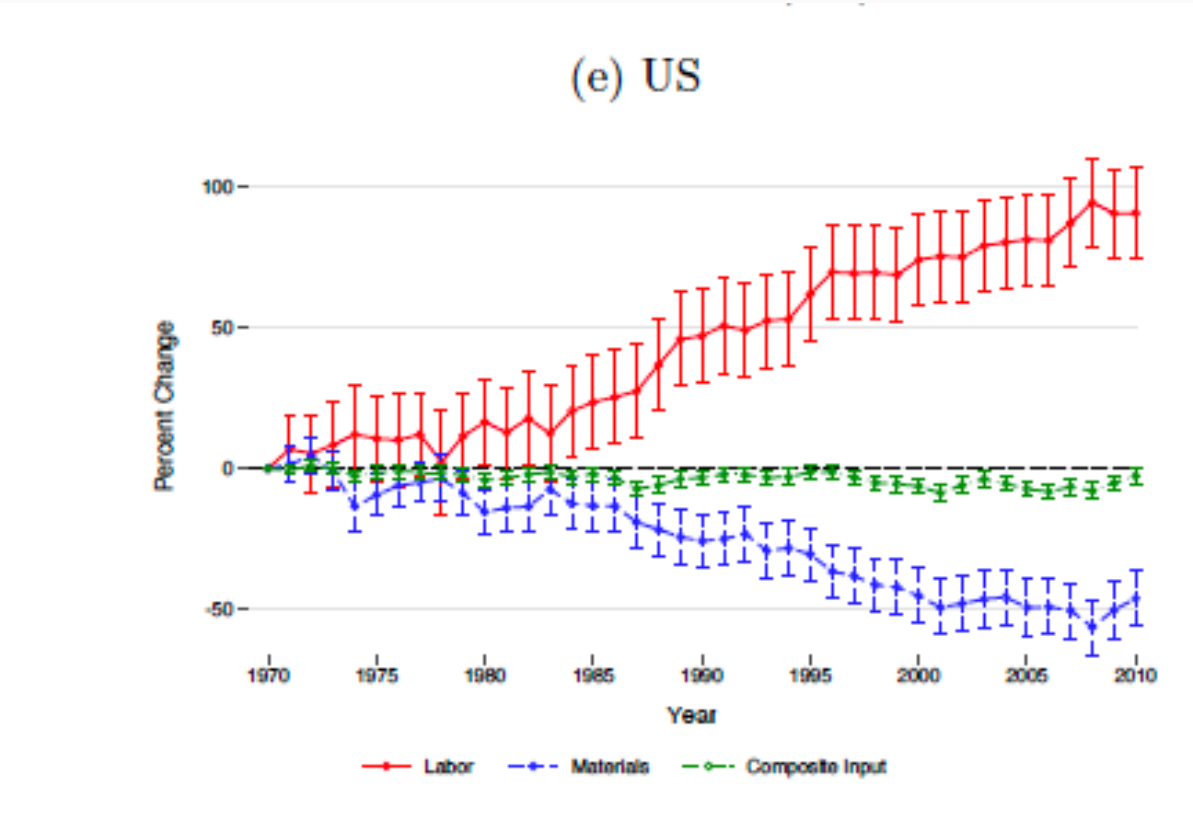
- Cost minimization gives $\lambda_{it} \frac{\partial F_{it}}{\partial (B_{it}L_{it})} B_{it} = W_{it}^{L}$ and $\lambda_{it} \frac{\partial F_{it}}{\partial M_{it}} = W_{it}^{M}$, so by definition

$$\frac{s_{it}^{L}}{s_{it}^{M}} = \frac{W_{it}^{L}L_{it}}{W_{it}^{M}M_{it}} = \frac{\frac{\partial F_{it}}{\partial (B_{it}L_{it})} B_{it} \frac{L_{it}}{Q_{it}}}{\frac{\partial F_{it}}{\partial M_{it}} \frac{M_{it}}{Q_{it}}}$$

- Intuitively B_{it} is *unobserved* in the data and adds one degree of freedom that reconciles output elasticities across inputs with their factor shares. Conversely, constructing the output elasticity from L_{it} alone attributes labor-augmenting technological change to markups (as in $\hat{\mu}_{it}^{L}$ in DEU).
- This is critique #1: a technology story that mimics a markup story.

Raval 2022: How Big Is the Bias?

- Backing out implied factor-augmenting technical change in US manufacturing: **labor**-augmenting productivity grows strongly, **material**-augmenting productivity declines.
- That is precisely the wedge that inflates $\hat{\mu}^L$ and deflates $\hat{\mu}^M$.



Demirer 2025: Taking the Technology Story Seriously

- Raval (2022) is a *diagnosis*. Demirer (2025) is the *treatment*: estimate a production function that *allows* labor-augmenting productivity, then recompute markups.
- Develops a new method to estimate production functions with *labor-augmenting* productivity.
- Establishes non-parametric identification and a novel control function approach.
- Fully flexible production function in capital, labor, and materials — output elasticities can be heterogeneous across firms.
- Uses the estimated production function to construct markups and an aggregate trend — directly comparable to DEU's headline series.

Demirer 2025: Technology

- A flexible production function (homothetically separable)

$$Y_{it} = F_t\left(\underbrace{K_{it}}_{\text{quasi-fixed}}, \underbrace{h_t(K_{it}, \omega_{it}^L L_{it}, M_{it})}_{\text{variable input}}\right) \exp(\omega_{it}^H) \exp(\epsilon_{it})$$

- This nests many familiar forms — Cobb-Douglas, CES, translog and more. $h_t(K_{it}, \cdot, \cdot)$ is homogeneous of arbitrary degree.
- Firms minimize cost given $K_{it}, \omega_{it}^H, \omega_{it}^L$, so the material-to-labor ratio can be written

$$\tilde{M}_{it} = r_t(K_{it}, \omega_{it}^L)$$

  In other words, it does *not* depend on the Hicks-neutral productivity shock ω^H.
- As long as the elasticity of substitution between materials and labor is not 1 (i.e. not Cobb-Douglas), $r_t(K_{it}, \omega_{it}^L)$ is monotonic in ω_{it}^L.
- Finally, the *level* of material choice is monotonic in ω_{it}^H

$$M_{it} = s_t(K_{it}, \omega_{it}^L, \omega_{it}^H)$$

- Two unobservables, two monotone choice margins — this is what makes a double control function possible.

Control Function: Recovering ω^L

- Markovian structure of the productivity shocks, represented as a rank ordering

$$\omega_{it}^L = g_1(\omega_{it-1}^L, \omega_{it-1}^H, u_{it}^1), \quad u_{it}^1 | \omega_{it-1}^L, \omega_{it-1}^H \sim U[0,1]$$

- Since both r_t and s_t are invertible, the material-to-labor ratio can be written

$$\tilde{M}_{it} = r_t(K_{it}, \omega_{it}^L) = r_t(K_{it}, g_1(\omega_{it-1}^L, \omega_{it-1}^H, u_{it}^1))$$
$$= r_t(K_{it}, g_1(\bar{r}_{t-1}(\tilde{M}_{it-1}, K_{it-1}), \bar{s}_{t-1}(\tilde{M}_{it-1}, M_{it-1}, K_{it-1}), u_{it}^1))$$
$$\equiv \tilde{r}_t(K_{it}, W_{it-1}, u_{it}^1) \quad W_{it-1} := \{K_{it-1}, M_{it-1}, L_{it-1}\}$$

- So conditional on W_{it-1}, K_{it}, all variation in $\tilde{M}_{it}$ is attributable to u_{it}^1 (conditionally independent of observables)
- It is non-parametrically identified by

$$u_{it}^1 = F_{\tilde{M}_{it}|W_{it-\mathbf{1}}, K_{it}}\left(\tilde{M}_{it} | W_{it-1}, K_{it}\right)$$

- With u_{it}^1 in hand, we can "control" for the unobserved labor-augmenting productivity

$$\omega_{it}^L = g_1(\bar{r}_{t-1}(\tilde{M}_{it-1}, K_{it-1}), \bar{s}_{t-1}(\tilde{M}_{it-1}, M_{it-1}, K_{it-1}), u_{it}^1) \equiv c_{1t}(W_{it-1}, u_{it}^1)$$

Control Function: Recovering ω^H

- Similarly define

$$\omega_{it}^H = g_2(\omega_{it-1}^L, \omega_{it-1}^H, u_{it}^1, u_{it}^2), \quad u_{it}^2 | \omega_{it-1}^L, \omega_{it-1}^H, u_{it}^1 \sim U[0,1]$$

- It is straightforward (as above) to use

$$M_{it} = \tilde{s}_t(K_{it}, W_{it-1}, u_{it}^1, u_{it}^2)$$

- u_{it}^2 is non-parametrically identified by

$$u_{it}^2 = F_{M_{it}|W_{it-\mathbf{1}}, K_{it}, u_{it}^{\mathbf{1}}}\left(M_{it} | W_{it-1}, K_{it}, u_{it}^1\right)$$

- With u_{it}^2 in hand, we can "control" for the unobserved Hicks-neutral productivity

$$\omega_{it}^H \equiv c_{2t}(W_{it-1}, u_{it}^1, u_{it}^2)$$

- Compare with ACF: the standard proxy inverts a *single* flexible input for a *single* unobservable. Here the *ratio* identifies ω^L and the *level* then identifies ω^H.

Identification and Estimation

- Interestingly, the author also shows that only the *sum* of labor and material elasticities ($\theta_{it}^M + \theta_{it}^L$) is identified without imposing optimizing behavior.
- Under cost minimization they can be separated by

$$\frac{\theta_{it}^M}{\theta_{it}^L} = \frac{s_{it}^M}{s_{it}^L}$$

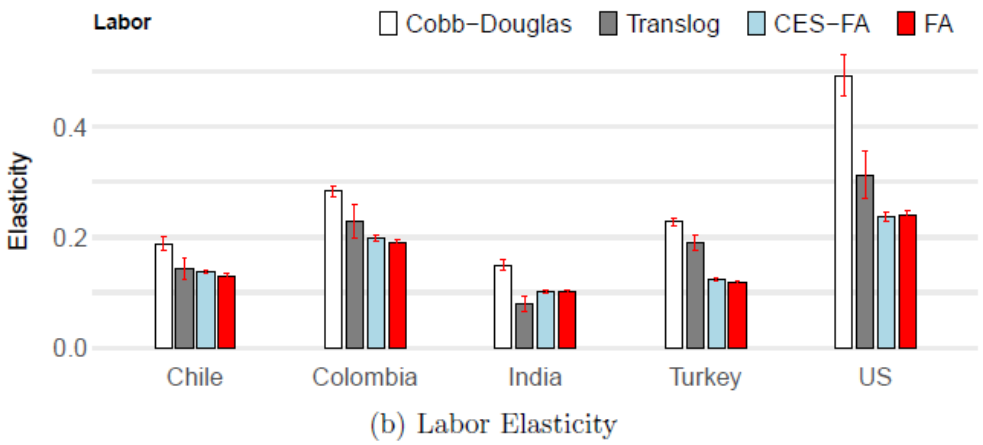
- For estimation, write h_t as a "reduced-form" function $L_t \bar{h}_t(\omega_{it}^L, \tilde{M}_{it}) \approx L_t \bar{h}_t(\tilde{M}_{it})$ (the author simplifies by taking K_{it} out of $h_t(\cdot)$)
- It is then natural to use the control-augmented production function to define moment restrictions

$$y_{it} = f_t(K_{it}, L_t \bar{h}_t(\tilde{M}_{it})) + c_{2t}(W_{it-1}, u_{it}^1, u_{it}^2) + \epsilon_{it}, \qquad E[\epsilon_{it} | W_{it}, W_{it-1}, u_{it}^1, u_{it}^2] = 0$$

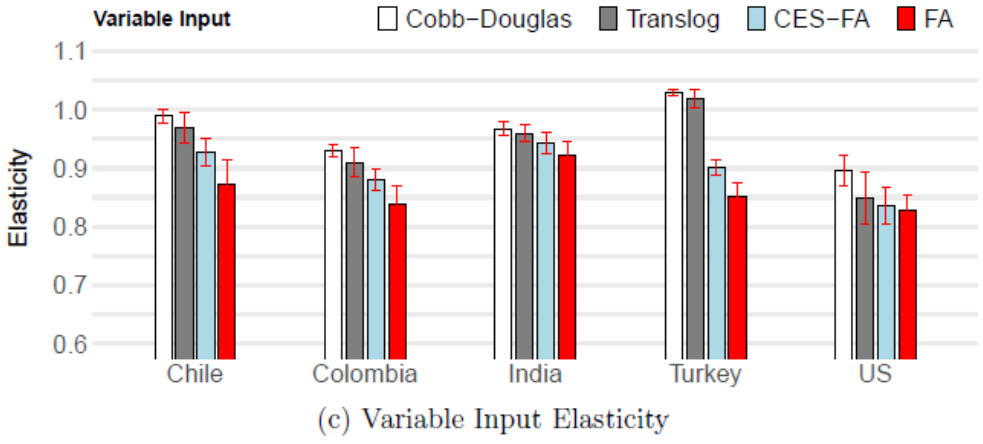
- Note: the restriction $\theta_{it}^M / \theta_{it}^L = s_{it}^M / s_{it}^L$ is exactly what Raval showed fails empirically under the standard approach — here it is *imposed* and ω^L absorbs the residual.

Demirer 2025: Output Elasticities Are Specification-Sensitive

- Labor output elasticity differs a lot across production specifications.



(b) Labor Elasticity

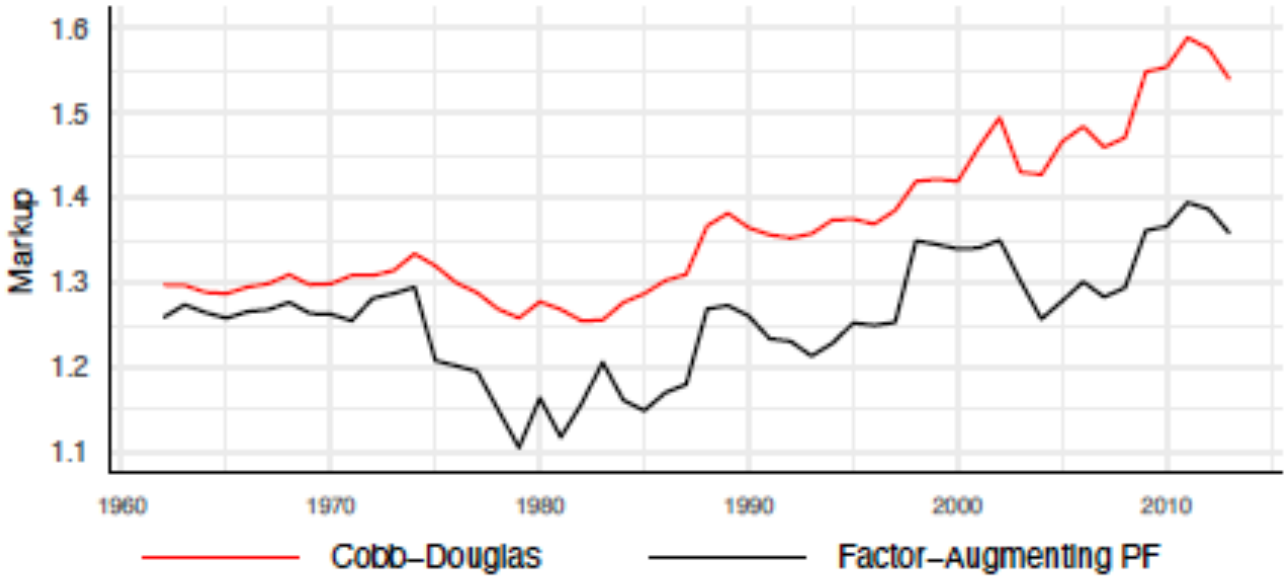


(c) Variable Input Elasticity

Demirer 2025: The Revised Aggregate Trend

- The aggregate sales-weighted markup rises to about $\mu = 1.4$, instead of 1.6.
- Read this carefully: the *level* is revised down, and so is the slope — but a positive trend survives. Technology explains part, not all, of DEU.

Figure 8: Sales-Weighted Markup (Compustat)



Notes: Comparisons of the evolution of markups in the US manufacturing industry produced by my method and the Cobb-Douglas production function.

Revenue vs. Quantity: Bond et al. 2021

- The production-based markup literature *assumes* both revenue and quantity are observed. In practice revenue is used for *both production function estimation and factor share calculation*.
- Production function estimation then suffers the well-known "omitted price bias" (Klette and Griliches, 1996).
- Bond et al. (2021) additionally highlight that the "constructed" markup uses a *revenue* elasticity rather than an output elasticity

$$\hat{\mu}_{it}^{R} = \frac{\theta_{it}^{R,j}}{s_{it}^{j}}$$

- Can μ_{it}^{R} be informative about μ_{it}? Bond et al. provide a dramatic example: a monopolist facing $P_{it} = \mathcal{P}(Q_{it})$.

Bond et al. 2021: A Degenerate Result

- Consider the monopoly firm's profit maximization problem

$$\max_{Q_{it}} \{P_{it}Q_{it} - C(Q_{it}; W_t)\} \quad s.t. \quad P_{it} = \mathcal{P}(Q_{it})$$

- The FOC gives the well-known formula $\mu_{it} = \frac{\eta_{it}}{\eta_{it}-1}$, where $\eta_{it} \equiv |\frac{P_{it}}{Q_{it}}\frac{\partial Q_{it}}{\partial P_{it}}| > 1$
- But the revenue elasticity is

$$\theta_{it}^{R,j} = \frac{X_{it}^j}{R_{it}}\frac{\partial R_{it}}{\partial X_{it}^j} = \frac{X_{it}}{Q_{it}}P_{it}^{-1}\left[\frac{\partial P_{it}}{\partial Q_{it}}\frac{\partial Q_{it}}{\partial X_{it}^j}Q_{it} + \frac{\partial Q_{it}}{\partial X_{it}^j}P_{it}\right] = \frac{X_{it}}{Q_{it}}\frac{\partial Q_{it}}{\partial X_{it}^j}\left[1 - \frac{1}{\eta_{it}}\right] \equiv \left(\frac{\eta_{it}-1}{\eta_{it}}\right)$$

- Substituting back,

$$\hat{\mu}_{it}^R = \frac{\theta_{it}^{R,j}}{s_{it}^j} = \left(\frac{\eta_{it}-1}{\eta_{it}}\right)\frac{\theta_{it}^j}{s_{it}^j} = \frac{1}{\mu_{it}}\mu_{it} = 1$$

- In other words, absent measurement error, the revenue-elasticity-based "markup" is degenerate at one and contains no information about the true markup at all.
- More generally, even without imposing profit maximization, the *level* of markups is not identified from revenue data alone. (This also motivates Demirer's robustness checks using quantity data.)

Demand Shocks: Bond et al. 2021

- A further challenge *even when quantity is observed* is unobserved demand shocks in variable input use.
- Consider a simple Cobb-Douglas technology (capital/labor quasi-fixed)

$$y_{it} = \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + (\omega_{it} + \epsilon_{it})$$

with downward-sloping demand $p_{it} = p(q_{it}, \xi_{it})$.

- Cost minimization in materials gives

$$m_{it} = \frac{\ln \beta_m}{1-\beta_m} + \frac{\beta_k}{1-\beta_m} k_{it} + \frac{\beta_l}{1-\beta_m} l_{it} + \frac{1}{1-\beta_m}(\underbrace{p_{it} - \ln \mu_{it}}_{\text{log marginal cost}} - p_{it}^M + \omega_{it})$$

- With a nonlinear productivity process $\omega_{it} = g(\omega_{it-1}) + \zeta_{it}$, we typically run a two-stage procedure (ACF 2015) that relies on a proxy function for ω_{it} in the first stage.
- But with variable markups the proxy $m_{it} = m^*(k_{it}, l_{it}, \omega_{it}, \xi_{it}, p_{it}^M)$ is *not invertible* for ω_{it}: two unobservables, one equation.
- Connection: this is the same scalar-unobservable problem Demirer solves for (ω^L, ω^H) — here the second unobservable is a *demand* shock, which the production approach has no instrument for.

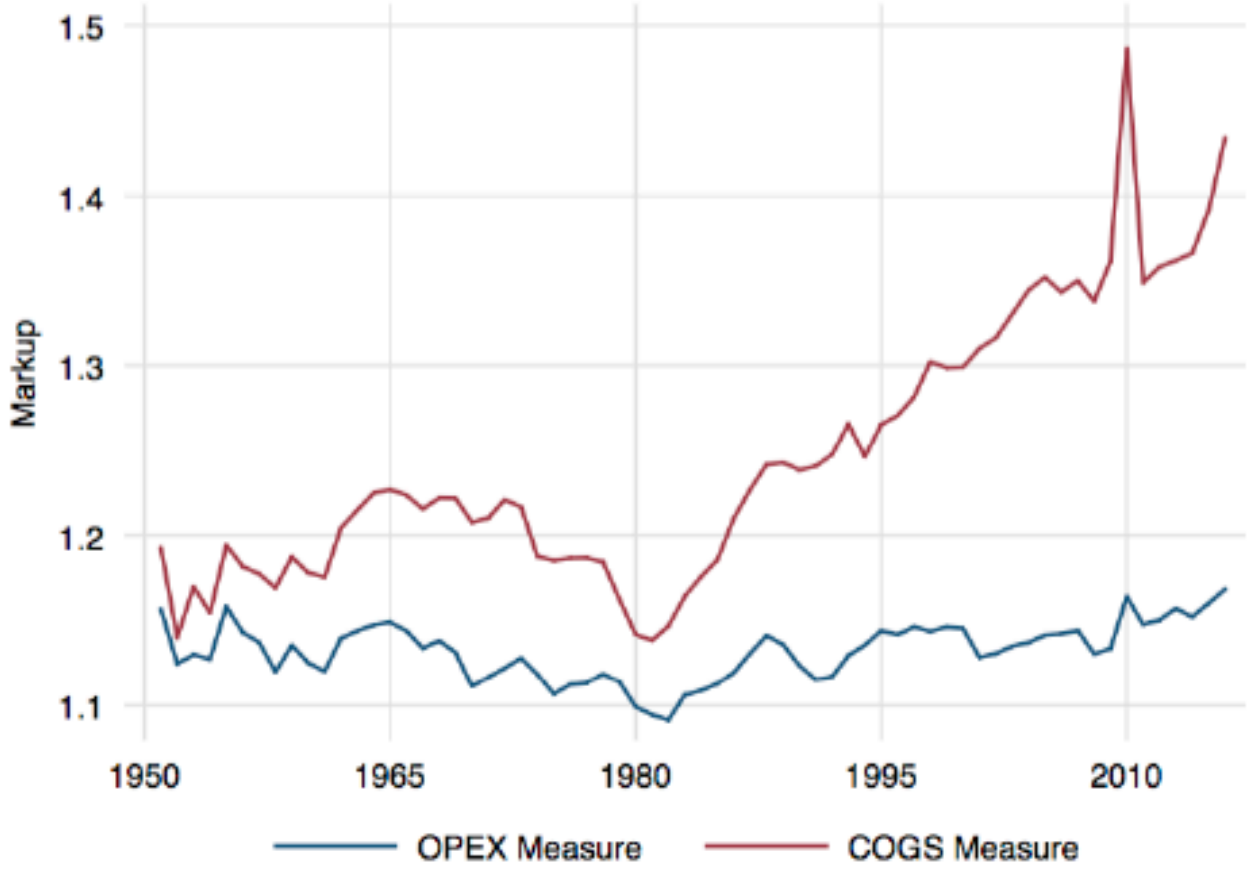
What Is Variable Cost: Traina 2018

- Traina (2018) makes an extremely simple point about Compustat (DEU's baseline): a financial statement may not separate variable from fixed cost well.
- The key debate is whether Selling, General and Administrative Expenses (SG&A) belong in variable cost.
- If they do, the markup measured on Cost of Goods Sold (COGS) + SG&A = OPEX shows little increase over time at all.
- The striking time-series variation is instead the *declining importance of COGS within OPEX* — see recent work emphasizing the rising importance of customer acquisition and marketing.
- Note how this differs from Raval/Demirer: they attack $\hat{\theta}^j$ (the numerator). Traina attacks s^j (the denominator). Both move $\hat{\mu} = \hat{\theta}^j / s^j$.

Traina 2018: Flat with OPEX

- The aggregate markup is essentially flat using OPEX.

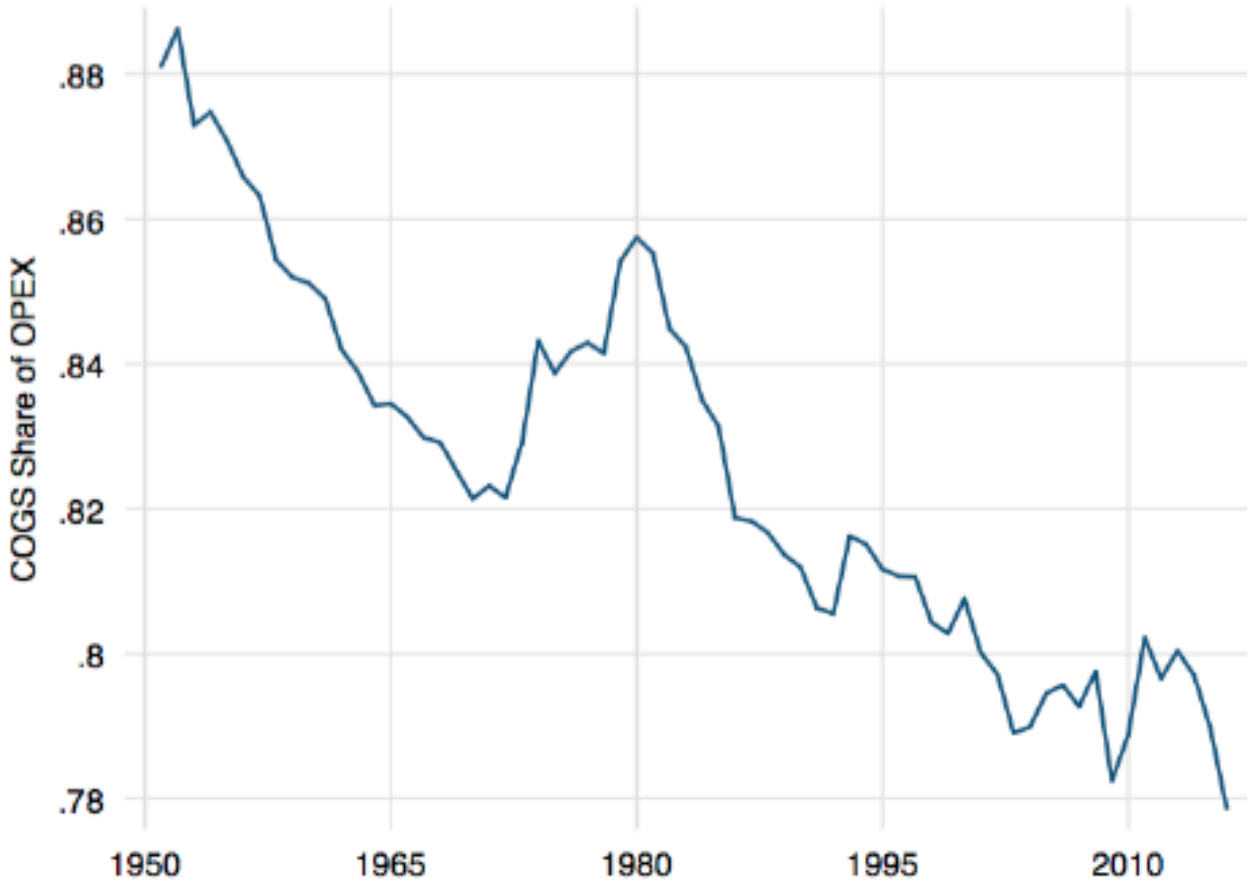
Figure 2: COGS vs. OPEX Markups



Traina 2018: The Composition of Cost Is What Moves

- COGS share declines steadily from 1980 to 2018.

Figure 4: COGS Share of OPEX



Scorecard: What Each Critique Does to DEU

Paper	Target	Level of μ	Trend in μ
Raval (2022)	$\hat{\theta}^j$: input-augmenting technology	—	attenuates
Demirer (2025)	$\hat{\theta}^j$: estimate ω^L directly	$1.6 \rightarrow 1.4$	attenuates, survives
Bond et al. (2021)	$\hat{\theta}^{R,j}$: revenue $\neq$ output	not identified	silent
Traina (2018)	s^j: COGS vs. OPEX	lower	eliminates

- No single critique is decisive; jointly they say the **level** is close to unidentified from Compustat and the **trend** is fragile to specification.
- They are also not independent: Traina's "rising SG&A" is a composition change that Raval's B_{it} would also pick up as biased technical change.
- What none of them can do: tell us whether consumers were made better or worse off. For that we need demand — which is Part II and Part III.

Aggregation and Welfare: Why We Care

- Suppose we *did* measure μ_{it} perfectly. What would we do with it? This is a simplified version of Edmond, Midrigan, and Xu (2023).
- Consider a firm with unit cost W/z_i, where z_i is firm productivity. (Think of labor as the only input, $y_i = z_i l_i$.)
- Profit maximization

$$\max_{p_i} \; p_i y_i - \frac{W}{z_i} y_i$$

  It is easy to show $p_i = \frac{\sigma_i}{\sigma_i - 1} \frac{W}{z_i} \equiv \mu_i \frac{W}{z_i}$, where the (endogenous) demand elasticity is $\sigma_i = -\frac{\partial \log y_i}{\partial \log p_i}$
- The input share is then trivially $\frac{W l_i}{p_i y_i} = \frac{1}{\mu_i}$. How do the *level* and the *distribution* of μ_i affect aggregate input use and efficiency?

Aggregation: What Is "The" Aggregate Markup?

- Given aggregate output Y and input $L = \int_i l_i$, define aggregate productivity as $Z = Y/L$. It is straightforward to show

$$Z = \left(\int_i \frac{1}{z_i} \frac{y_i}{Y} \right)^{-1}$$

- Similarly define the aggregate markup as $\mu = P/(W/Z)$. Substituting in Z,

$$\mu = (PY)/(WL)$$

- Using $\frac{p_i y_i}{PY} = \frac{\mu_i}{\mu} \frac{l_i}{L}$ and integrating both sides,

$$\mu = \int_i \mu_i \frac{l_i}{L} \equiv \left(\int_i \frac{1}{\mu_i} \frac{p_i y_i}{PY} \right)^{-1}$$

  The aggregate markup is an *input-weighted arithmetic average*, equivalently a *sales-weighted harmonic average*, of firm-level markups.

- This matters for reading DEU: their headline series is sales-weighted *arithmetic*. The weighting scheme is not innocuous when the markup distribution has a fat right tail.

Aggregation: Level vs. Dispersion

- The aggregate labor share $\frac{1}{\mu} = (WL)/(PY)$ can fall for two reasons: (1) $\frac{1}{\mu_i}$ falls; (2) the sales share $\frac{p_i y_i}{PY}$ rises more for low-$\frac{1}{\mu_i}$ firms.
- For allocative efficiency we need an assumption on how firm outputs aggregate. Let $Y = \left(\int_i y_i^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}$, which implies

$$y_i/Y = (p_i/P)^{-\theta} \equiv \left(\frac{\mu_i}{\mu} \frac{Z}{z_i} \right)^{-\theta}$$

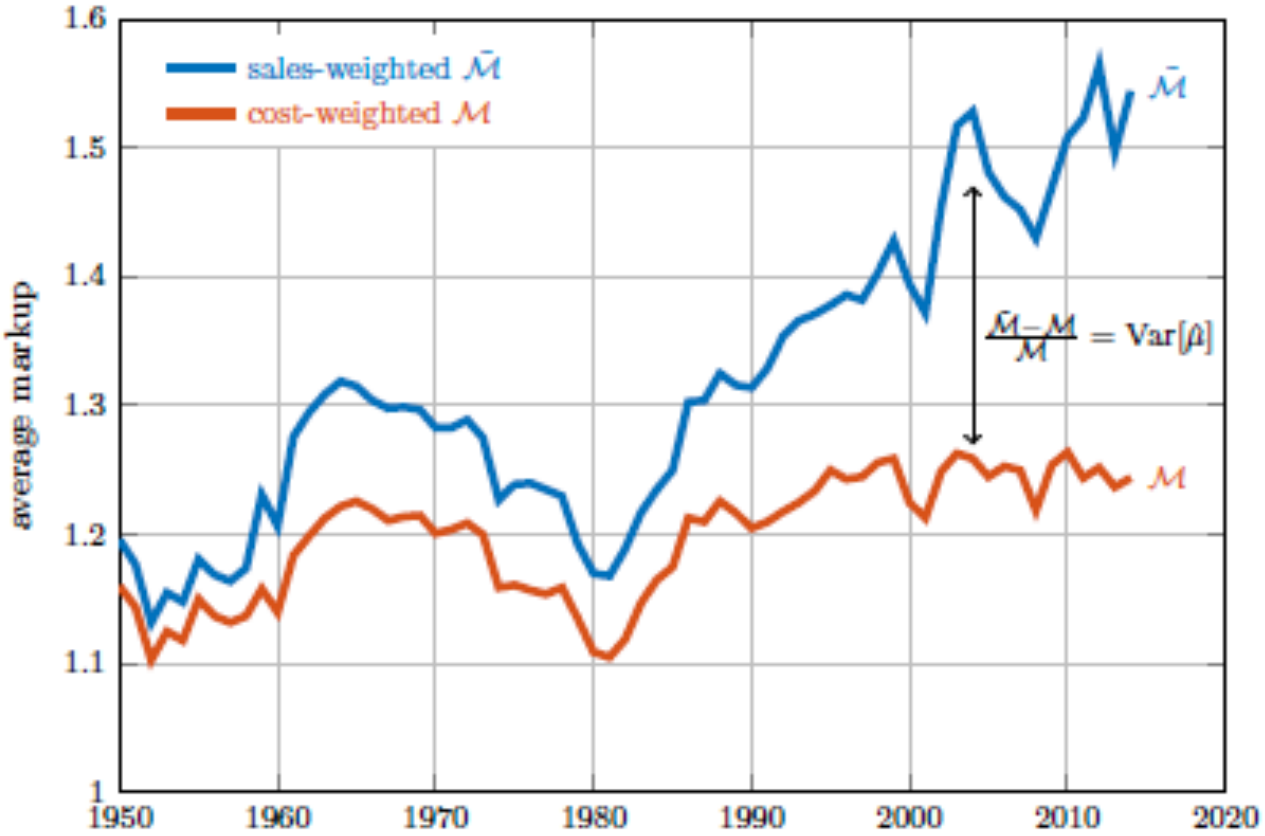
- Plugging back into the expression for Z and solving,

$$Z = \left(\int_i \left(\frac{\mu_i}{\mu} \right)^{-\theta} z_i^{\theta-1} \right)^{\frac{1}{\theta-1}}$$

- Key result: only *relative* markups μ_i/μ enter. A uniform rise in the markup level is a pure transfer; it is rising **dispersion** that reduces aggregate efficiency Z.
- This reframes the empirical question: not "did μ rise?" but "did the *spread* of μ_i rise?" — a much harder object to measure, and much more sensitive to every critique in Part I.

Edmond, Midrigan, and Xu 2023

Figure A1: Cost-Weighted vs. Sales-Weighted Average Markups, Compustat



The sales-weighted average $\tilde{\mathcal{M}}$ of firm-level markups in Compustat data, as in De Loecker, Eeckhout and Unger (2020), and the cost-weighted average of firm-level markups $\mathcal{M}$. The former is higher and has increased by a larger amount. The proportional difference between the two averages reflects the cross-sectional dispersion in markups, which has been increasing.

Why Go to a Single Industry?

The aggregate literature has hit a wall on three fronts. Each is something an industry study can do and a Compustat panel cannot:

1. **Marginal cost.** The production approach infers *mc* from a technology restriction. IO infers it from a demand system plus a conduct assumption. These are different identifying assumptions, so they are a genuine cross-check.
2. **Product quality and variety.** Over 40 years a "car" is not the same good. A markup series is silent on whether the *product* improved — yet that is most of the welfare action.
3. **Conduct.** The production approach has *no* conduct parameter: μ is whatever the technology residual says. It cannot tell a merger that raises elasticity-driven markups from one that facilitates coordination.

- Case study 1 (autos) takes on (1) and (2): same industry as DEU, opposite answer, and a full welfare decomposition.
- Case study 2 (beer) takes on (3): markups move because *conduct* moves, which no technology-based estimator can see.

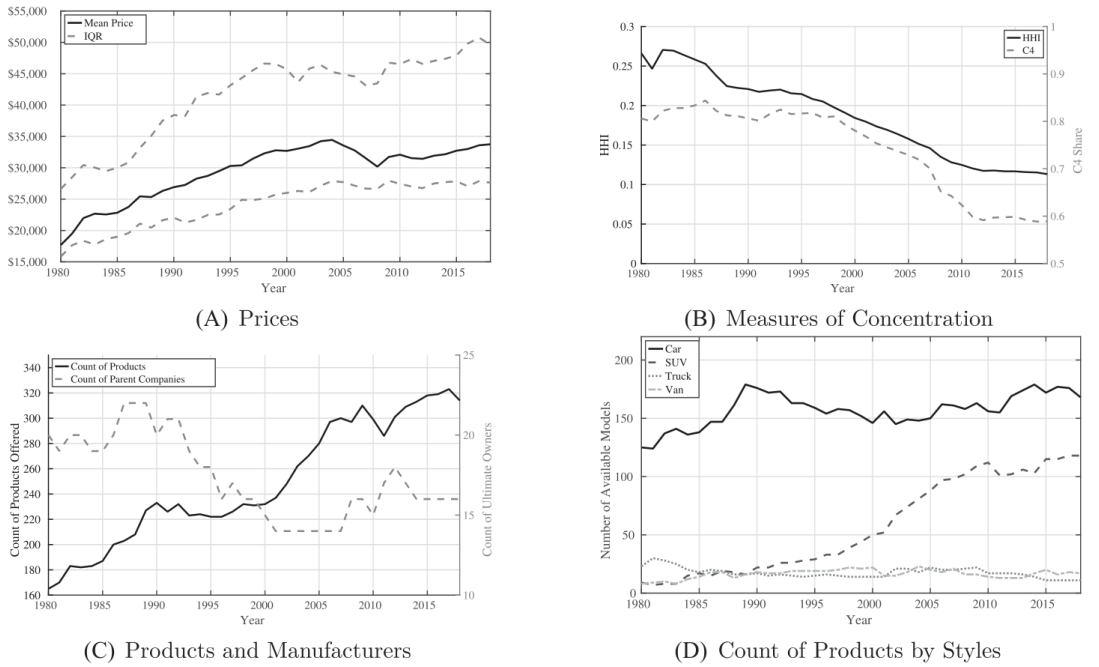
Case Study 1: Grieco, Murry, and Yurukoglu (2024, QJE)

- Debate in macro vs. IO: are markups rising? Are markets less competitive?
- Autos: a large share of durable goods spending, a long horizon (1980–2018), and the industry BLP (1995) made the canonical IO laboratory.
- An ideal setting to compare:
  - **Aggregate approach**: production-based accounting markups (DEU).
  - **Industry-specific IO**: demand estimation plus a Nash-Bertrand supply side.
- Key questions:
  1. How did prices, concentration, and product variety evolve?
  2. Did markups rise or fall?
  3. What drove changes in consumer welfare?
- Preview: real prices roughly doubled, median Lerner index *halved*, and the consumer share of surplus rose from 0.62 to 0.82.

Data Sources

- **Ward's Automotive** (1980–2018): sales, MSRP, characteristics (horsepower, MPG, footprint, height, curb weight).
- **CEX (1980–2005) / MRI (1992–2018)**: demographics linked to purchases; identifies heterogeneity in α_i, β_i.
- **MaritzCX**: stated "second choices" (1991, 1999, 2005, 2015; $N \approx$ 29k, 20k, 43k, 53k) for diversion micro-moments.
- **RXR instrument**: production-country real exchange rate (PPP-adjusted), Penn World Tables 9.1, lagged one year.
- Market size: households/2.5 (roughly two cars per household).
- Why this data combination: aggregate shares alone pin down own-price elasticities poorly over a 39-year panel. The micro-moments discipline *substitution*, which is what markups actually depend on.

Prices & Market Structure, 1980–2018



(A) Prices

(B) Measures of Concentration

(C) Products and Manufacturers

(D) Count of Products by Styles

FIGURE II

Prices and Market Structure, 1980–2018

- Real average MSRP roughly doubles ($\approx$ \$17,000 $\rightarrow\approx$ \$34,000 in 2015 USD).
- Concentration *falls*: parent-level HHI from above 2500 to about 1200; C4 from ≈ 0.80 to ≈ 0.58.
- Number of models roughly doubles (165 in 1980 to over 314 in 2018).

Physical Characteristics, 1980–2018

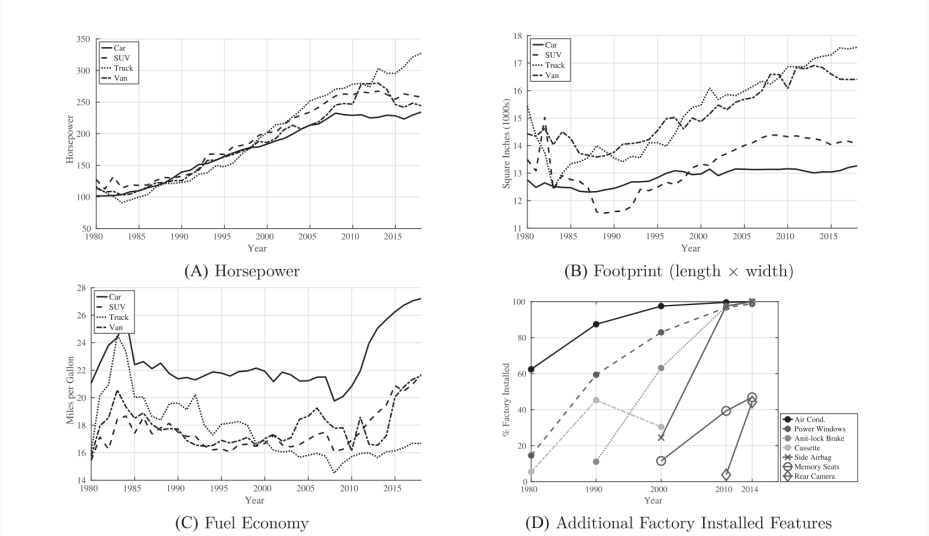


FIGURE III

Physical Vehicle Characteristics, 1980–2018

- SUVs rise from near zero to roughly half of sales; vehicles become larger and far more powerful (car horsepower roughly doubles).
- MPG roughly stable — efficiency gains absorbed by weight and power under CAFE standards.
- Safety and infotainment features diffuse: A/C 61% $\rightarrow$ 99%, power windows 15% $\rightarrow$ 99%, ABS 0% $\rightarrow$ 80%.
- The identification problem in one slide: prices doubled, but so did the product. A price series is uninformative about welfare without a quality adjustment.

Demand: Consumer Utility

- Random coefficients discrete choice:

$$u_{ijt} = \beta_{it}' x_{jt} + \alpha_{it} p_{jt} + \xi_{jt} + \epsilon_{ijt}, \qquad u_{i0t} = \gamma_t + \epsilon_{i0t}$$

- Heterogeneity in both observed demographics and unobserved taste:

$$\alpha_{it} = \bar{\alpha} + \sum_h \alpha_h D_{it}^h, \qquad \beta_{ikt} = \bar{\beta}_k + \sum_h \beta_{kh} D_{it}^h + \sigma_k \nu_{ik}$$

  with D_{it} = income, income^2, age, rural, family size; $\nu_{ik} \sim N(0,1)$.

- Micro-moments come in two flavors:
  - **Demographic moments** (CEX/MRI): predicted vs. observed average price paid by income quintile; vehicle type by family size and rural status.
  - **Second-choice moments** (MaritzCX): correlation of characteristics between the vehicle purchased and the stated second choice — these discipline *diversion*, following BLP (2004) and Petrin (2002).
- Why second choices matter here: the largest random coefficients are on *vehicle style*. Without them, a logit-style model over-rewards product proliferation — GMY report welfare gains roughly five times larger under logit.

Identification I: Quality vs. the Outside Option

- **The problem**: discrete choice identifies utility only *relative to the outside good*. Write $\xi_{jt} = \tau_t + \tilde{\xi}_{jt}$, where τ_t is mean unobserved quality. Market shares identify only the difference $\tau_t - \gamma_t$.
- Over 39 years this is fatal: a rising inside share is equally consistent with better cars ($\tau_t \uparrow$) or a worse outside option ($\gamma_t \downarrow$) — and these have opposite welfare readings.
- **The restriction** (Pakes, Berry, and Levinsohn 1993). Let C_t be the set of *continuing* vehicles whose design generation is unchanged from $t-1$:

$$\forall j \in C_t: \quad E\left[\xi_{jt} - \xi_{j,t-1}\right] = E\left[(\tau_t - \tau_{t-1}) + (\tilde{\xi}_{jt} - \tilde{\xi}_{j,t-1})\right] = 0$$

- **Why it works**: a model within a design generation is substantively the same car year to year. So any *average* movement in ξ across continuing products must be $\tau_t - \tau_{t-1}$. With $\tau_0 = 0$, τ_t is identified by chaining; γ_t then follows from $\tau_t - \gamma_t$.
- **External validation**: Kelley Blue Book resale values of mint-condition used cars. Year-of-production dummies rise by \$19,638.88 between 1992 and 2017 — close to the model's τ_t path, which is estimated from *new*-car shares only.

Quality vs. Outside Option (Time Effects)

- τ_t (mean unobserved quality) rises steadily and roughly linearly, by about \$25,000 cumulatively.
- γ_t (outside good) spikes in recessions (1990–91, 2007–09) — a business-cycle object, not a technology one.

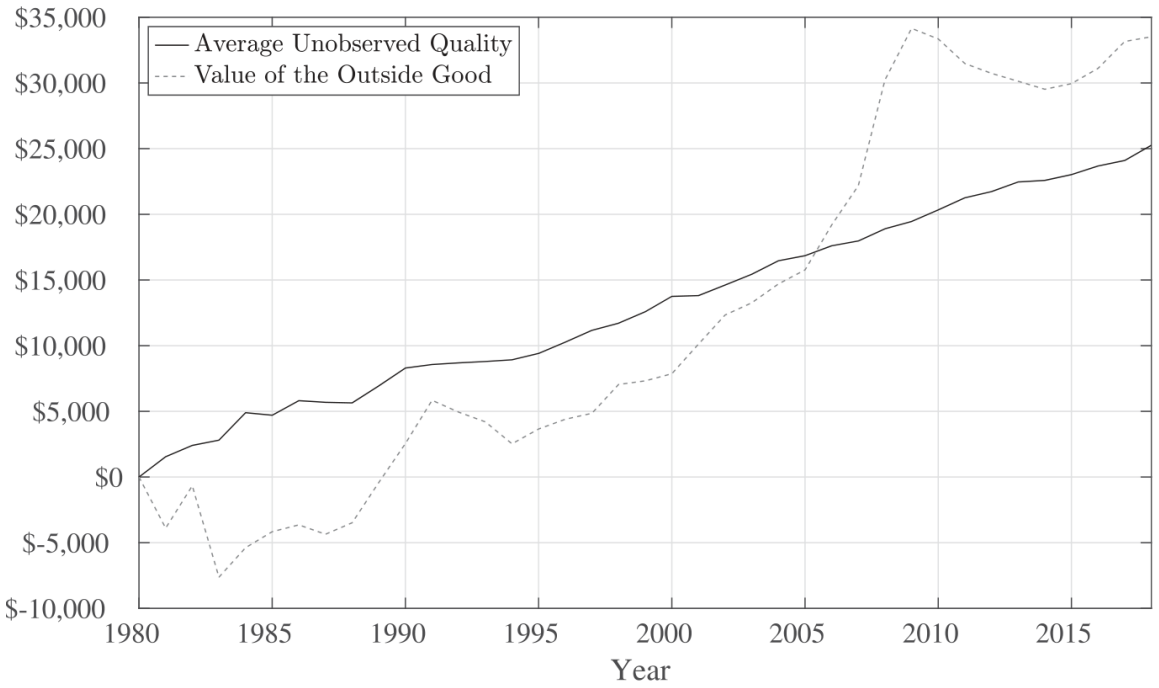


FIGURE V

Quality and Aggregate Components of Time Effects

Identification II: Prices and the Real Exchange Rate

- Price is endogenous to ξ_{jt}, so we need a cost shifter. GMY use the **PPP-adjusted real exchange rate** of the vehicle's *production country*, lagged one year.
- Two channels: local wage and input costs enter through PPP; nominal exchange rate movements change the optimal markup of a foreign producer selling in USD.
- Crucially, there is variation *within make* both across models and over time — BMW assembles in Germany and the US, GM in Canada, Mexico, and South Korea.
- First stage: coefficient 0.411 (0.110), $F = 14.09$, implying a pass-through of RXR to prices of 0.117 — consistent with the exchange-rate pass-through literature.
- **OLS vs. IV** shows large, correctly signed endogeneity bias:

	OLS	IV
Price coefficient	−0.334	−1.696
Mean own-price elasticity	−1.204	−6.107

- Sanity check: the OLS elasticity of −1.2 is barely above the monopoly bound. Uncorrected, it would deliver enormous markups — in the same direction as the accounting approach.

Supply: Markups and Marginal Costs

- Manufacturer m prices all vehicles in $\mathcal{J}_{mt}$ under static multi-product Nash-Bertrand with constant marginal cost c_{jt}:

$$s_{jt} + \sum_{k \in \mathcal{J}_{mt}} (p_{kt} - c_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0$$

- Stacking and inverting recovers product-level marginal cost:

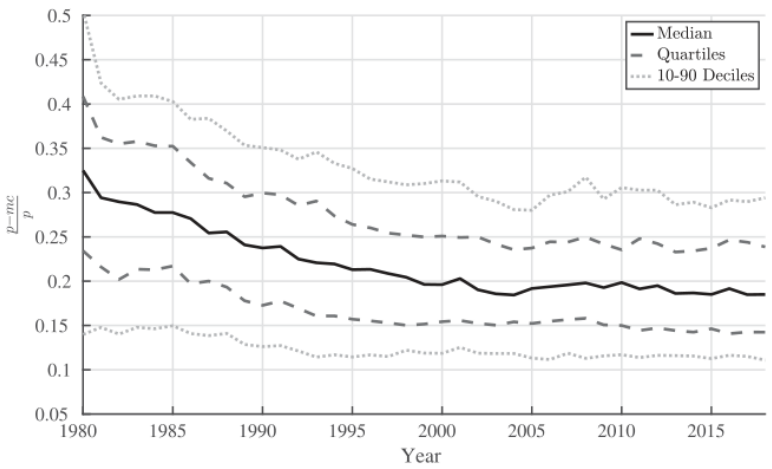
$$c_t = p_t + \left[\Omega_t \circ \left(\frac{\partial s_t}{\partial p_t} \right)^T \right]^{-1} s_t$$

  where Ω_t is the ownership matrix. This is exactly the object we will let κ move in the beer case.

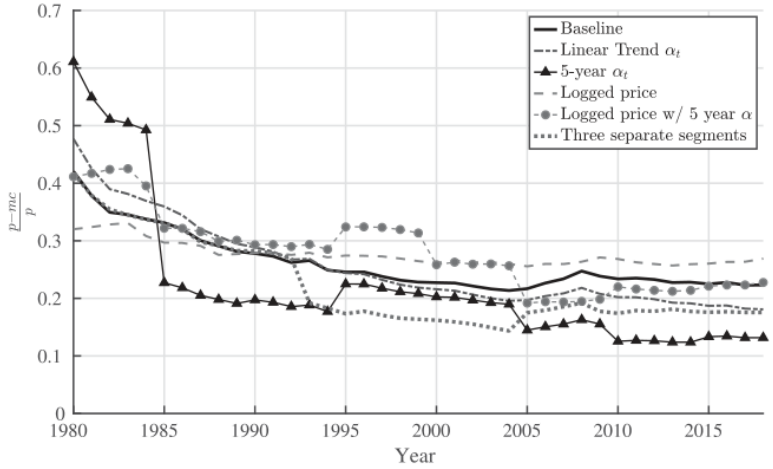
- Report the **Lerner index** $\mathcal{L}_{jt} = (p_{jt} - c_{jt})/p_{jt}$, equivalently $\mu_{jt} = p_{jt}/c_{jt} = 1/(1 - \mathcal{L}_{jt})$ — the same object DEU estimate.
- Maintained assumption: static Nash-Bertrand *every* year. This rules out changes in conduct over the period; robustness to alternative conduct is checked in Section 5.5.
- Note the trade: GMY buy quality and welfare measurement at the price of *assuming* conduct is constant. Miller and Weinberg will buy conduct at the price of a much narrower setting.

Markups Over Time

- Median Lerner index falls from 0.325 (1980) to 0.185 (2018); the share-weighted mean falls from about 0.42 to 0.22.
- Equivalently $\mu = p/mc$ falls from about 1.78 to 1.29. Markups fall *across the whole distribution*, not just at the median.
- Robust to a linear trend in α_t, to 5-year segments for α_t, to log-price utility, and to alternative conduct assumptions.



(A) Distribution of Markups



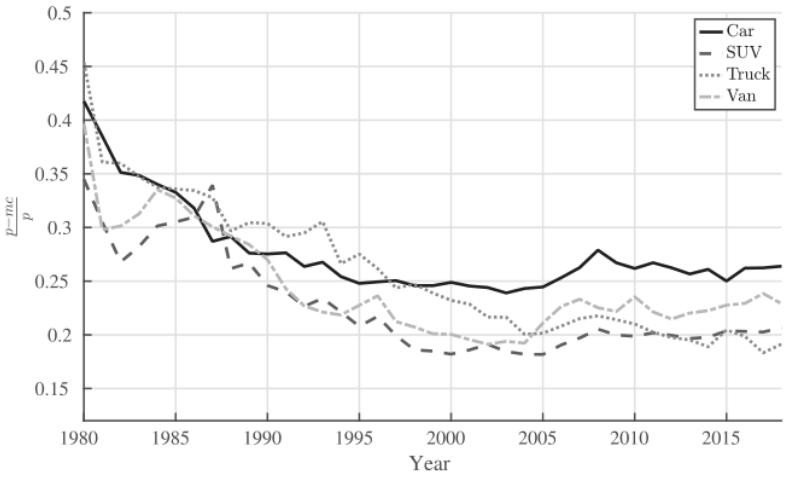
(B) Average Markups, Additional Specifications

FIGURE VI

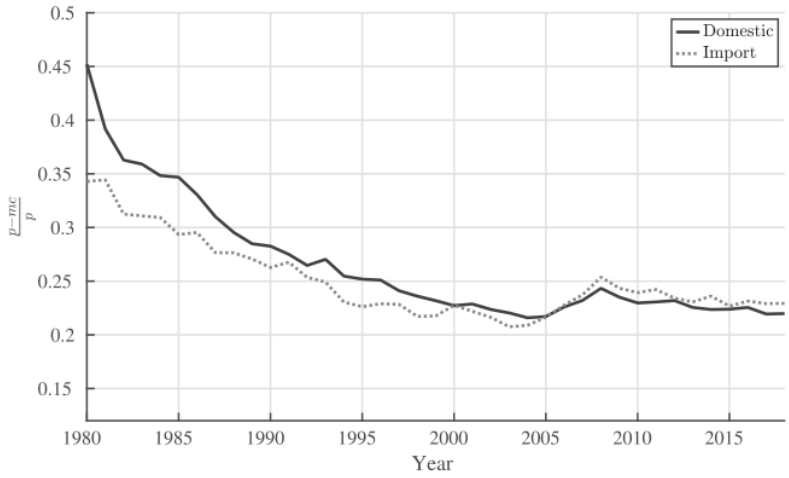
Markups

Markups by Vehicle Style / Origin

- The decline is not composition: trucks and SUVs fall steeply (especially in the 1990s), cars fall steadily, and domestic and imported markups *converge*.



(A) Markups by Vehicle Style



(B) Markups by Import Status

FIGURE VII

Why Do Markups Fall While Prices Rise?

The two facts are not in tension once you hold the product fixed:

1. **Quality raises marginal cost.** Bigger, more powerful, more feature-laden vehicles cost more to build. Estimated c_{jt} trends up over the sample.
2. **Pass-through is incomplete.** A differentiated-product oligopolist passes through less than one-for-one, so $p_{jt} - c_{jt}$ rises by *less* than c_{jt} does. Mechanically $\mathcal{L}_{jt} = 1 - c_{jt}/p_{jt}$ falls.
3. **Competition intensified.** Import entry and product proliferation raised own-price elasticities (mean elasticity moves from about -4.2 in 1985 to -5.4 in 2015) — the opposite of the "superstar firm" narrative.

- **Counterfactual test** (Figure IX): rescale each year's characteristic distribution to the 1980 percentile distribution. Marginal costs then show essentially *no* upward trend, and the markup decline is substantially attenuated.
- Why costs are flat in the counterfactual despite estimated technical progress: there is an offsetting upward trend in the RXR over the sample.

Welfare: Measuring Consumer Surplus over 39 Years

- With quasi-linear utility, compensating variation for consumer i in year t is the standard log-sum,

$$CS_{it}(\gamma) = \frac{1}{|\alpha_{it}|}\left[\log\left(\exp(\gamma) + \sum_{j\in\mathcal{J}_t}\exp(\beta_{it}'x_{jt} + \alpha_{it}p_{jt} + \xi_{jt})\right)\right]$$

- **Problem**: evaluating this at the year-t outside good γ_t mixes product improvements with business-cycle swings in the value of *not* buying. A recession then looks like a welfare loss from the car market.
- **Fix**: hold the outside option fixed at its average across the sample,

$$\overline{CS}_t = \frac{1}{T}\sum_{v=0}^{T} CS_t(\gamma_v)$$

  so that movements reflect only the evolving *product set* and prices.
- Report the consumer share of **efficient surplus** $= CS + PS + DWL$: this is scale-free and does not require CS's additive constant.
- Contrast with Part I: this is the question the aggregate literature cannot ask. A markup number alone cannot sign the welfare change.

Consumer Surplus, Producer Surplus, DWL

- *CS* rises by roughly \$8,000 per household; the consumer share of efficient surplus rises from 0.62 to 0.82.
- *PS* rises much more slowly; *DWL* stays small relative to total surplus.

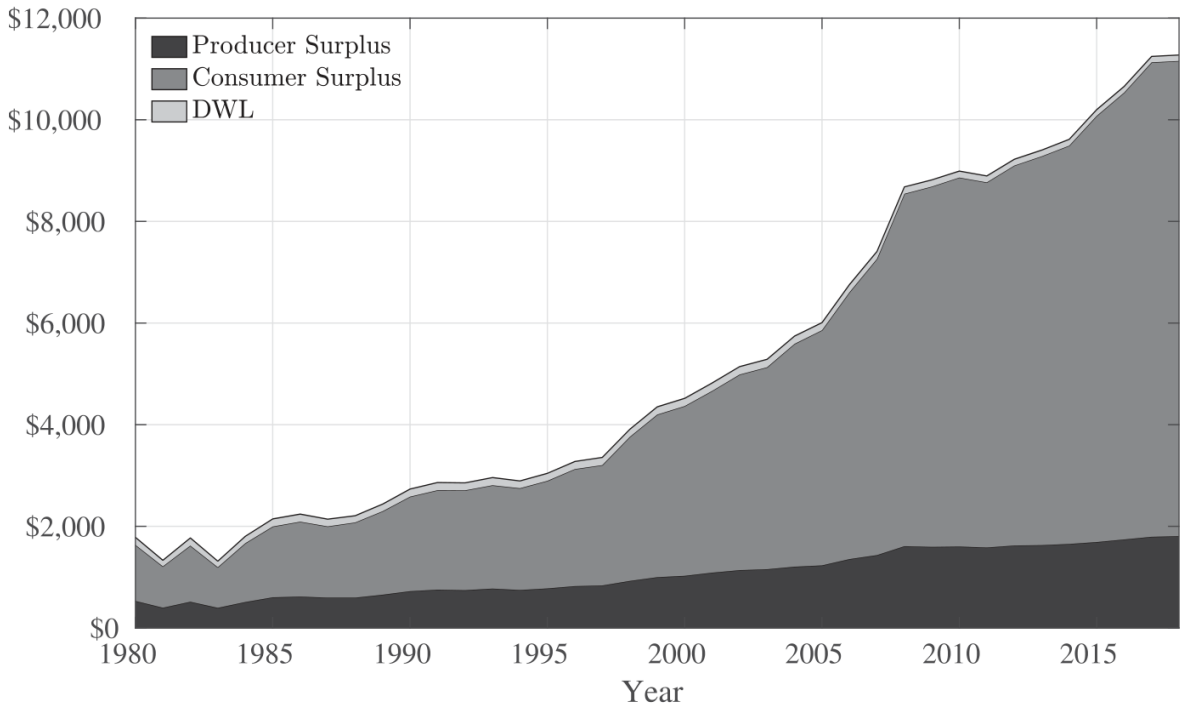


FIGURE XII

Consumer Surplus, Producer Surplus, and Deadweight Loss

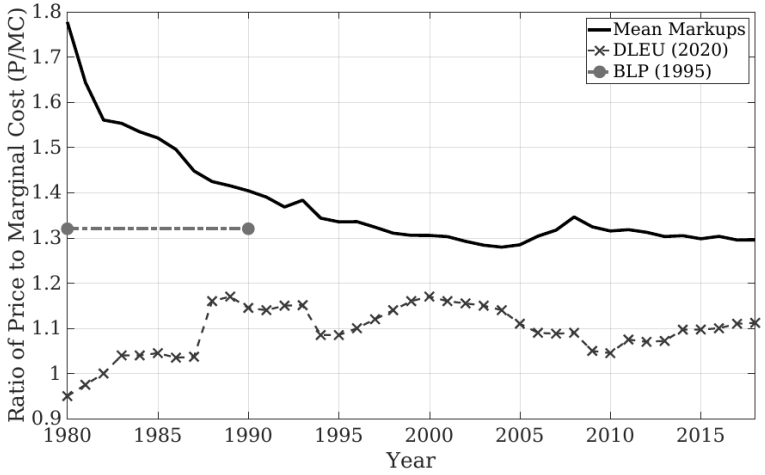
Counterfactuals: What Actually Drove the Welfare Gain?

Four mechanisms, each shut down in turn:

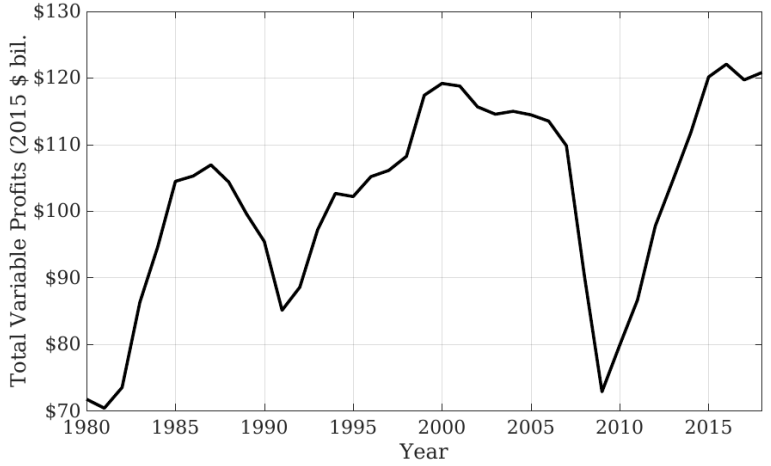
1. **Import competition.** Reassign all foreign-brand vehicles to the Big 3 (averaged over 10 random draws). *CS* still increases substantially $\Rightarrow$ foreign entry is *not* the main driver. Benchmarks: Big 3 collusion, and a full industry cartel, dampen but do not eliminate the gain.
2. **Product proliferation.** Hold the choice set at the 1980 level of 165 products (vs. 314 in 2018). Small gap from baseline — consumers substitute to close neighbors. Removing SUVs entirely also has only a modest effect.
3. **Observable attributes.** Rescale horsepower, MPG, footprint, height, curb weight to the 1980 mean and variance, holding *mc* fixed. *Modest* impact on *CS*.
4. **Unobserved quality and production technology.** Set $\tau_t = 0 \; \forall t$, and separately remove the estimated cost-side technical progress. **Largest** effects by far.

- Punchline: the welfare gain is a *technology* story — better cars and cheaper production — not a market-structure story. Note how badly this is served by the markup series alone.

Reconciliation: GMY vs. De Loecker et al. (2020)



(a) Price over Marginal Cost



(b) Total Variable Profits

- Panel (a): GMY's share-weighted p/mc falls from 1.78 to about 1.29; DEU's auto-sector series *rises* from about 0.95 to 1.11. BLP's (1995) 1971–90 average of 1.32 sits on GMY's path.
- Panel (b): GMY's *total variable profits* (margin $\times$ quantity) do track DEU's cyclical pattern — suggesting DEU's series is partly picking up **quantity** movements against fixed costs.

Why the Two Series Disagree

- **COGS is not a flexible input.** GAAP requires allocating fixed overhead into COGS. Ford's 1993 annual report: the industry "operates with a relatively high percentage of fixed costs" — yet Compustat records all of Ford's operating expenses that year as COGS. Including fixed cost in the flexible input generates spurious time variation in $\hat{\mu}$ driven purely by fluctuating quantity. *[Traina's critique, in one industry]*
- **Output elasticities are estimated off the wrong object.** Compustat reports firm-year *revenue*, not quantity *[Bond et al.]*; autos are pooled with computers, shipbuilding, and furniture at the 2-digit NAICS level; and multi-product production is ignored. *[Raval, Demirer]*
- **Sample coverage.** Only US-listed firms: VW enters only in 2001, Nissan in 1989, BMW never; Hyundai-Kia, Mazda, Mitsubishi largely absent. Revenue includes parts and financing, and *worldwide* sales at different markups. Daimler's 1988 entry (with large aerospace revenue) shifts the DEU series up.
- **Independent corroboration.** De Loecker and Scott (2022) find the two approaches *do* align for beer; Foster et al. (2022), using the Census of Manufactures, find decreasing manufacturing markups 1977–2012 in some specifications.

Case Study 1: What to Take Away

- Despite rising prices, markups fall and consumer surplus rises. Concentration falls too.
- The drivers are quality growth and production efficiency — *not* changes in market structure or conduct.
- Methodologically: industry-specific IO and aggregate accounting can deliver **opposite signs on the same industry**, and the difference is traceable to the assumptions we catalogued in Part I.
- The remaining gap: GMY *assume* static Nash-Bertrand throughout. If conduct itself changed, their markup series would absorb it into demand elasticities or costs. That is the question Case Study 2 takes up.

Case Study 2: When Conduct Is the Moving Part

- In autos, the ownership matrix Ω_t was given by legal ownership and *conduct was assumed*: static Nash-Bertrand in every one of 39 years. Antitrust practice is built on the opposite worry.
- The Horizontal Merger Guidelines distinguish:
  - **Unilateral effects**: the merged firm internalizes diversion between its own products $\Rightarrow$ a *new* Bertrand equilibrium, same equilibrium concept.
  - **Coordinated effects**: the merger softens competition between the merged firm and its *remaining rivals* $\Rightarrow$ the equilibrium concept itself changes.
- Essentially the entire empirical merger literature simulates the first and treats the second as unmeasurable — Deneckere and Davidson (1985), Hausman-Leonard-Zona (1994), Werden-Froeb (1994), Nevo (2000), Jaffe and Weyl (2013) all hold conduct fixed.
- Miller and Weinberg (2017) is the paper that puts a number on the second, using the 2008 MillerCoors joint venture.
- Why it belongs in *this* lecture: it is a market-power measurement exercise in which the markup rises **without** any change in technology, costs, or demand elasticities. Nothing in the production approach of Part I could detect it.

Miller and Weinberg 2017: Overview

- **Setting**: the 2008 Miller + Coors joint venture (US operations only). Coalition candidates are ABI and MillerCoors; the competitive fringe is Modelo and Heineken.
- **Data**: IRI scanner data, 2001–2011, 39 regions, 13 flagship brands $\times$ 3 package sizes; 94,656 brand-size-region-month observations. June 2008–May 2009 dropped as a transition window.
- **Facts to explain**: retail prices of *both* MillerCoors and ABI jump just after consummation; import prices continue on trend.
- **Strategy**: within a differentiated-products pricing model, *test and reject* the hypothesis that the price increases represent movement from one Nash-Bertrand equilibrium to another.
- **Results**: ABI and MillerCoors internalize about 26% of their pricing externalities. Post-merger prices are 6–8% higher and markups 17–18% higher than under Nash-Bertrand.
- Note the logic: this is an *ex post* evaluation, which is what lets conduct be estimated rather than assumed. The merger is the "experiment."

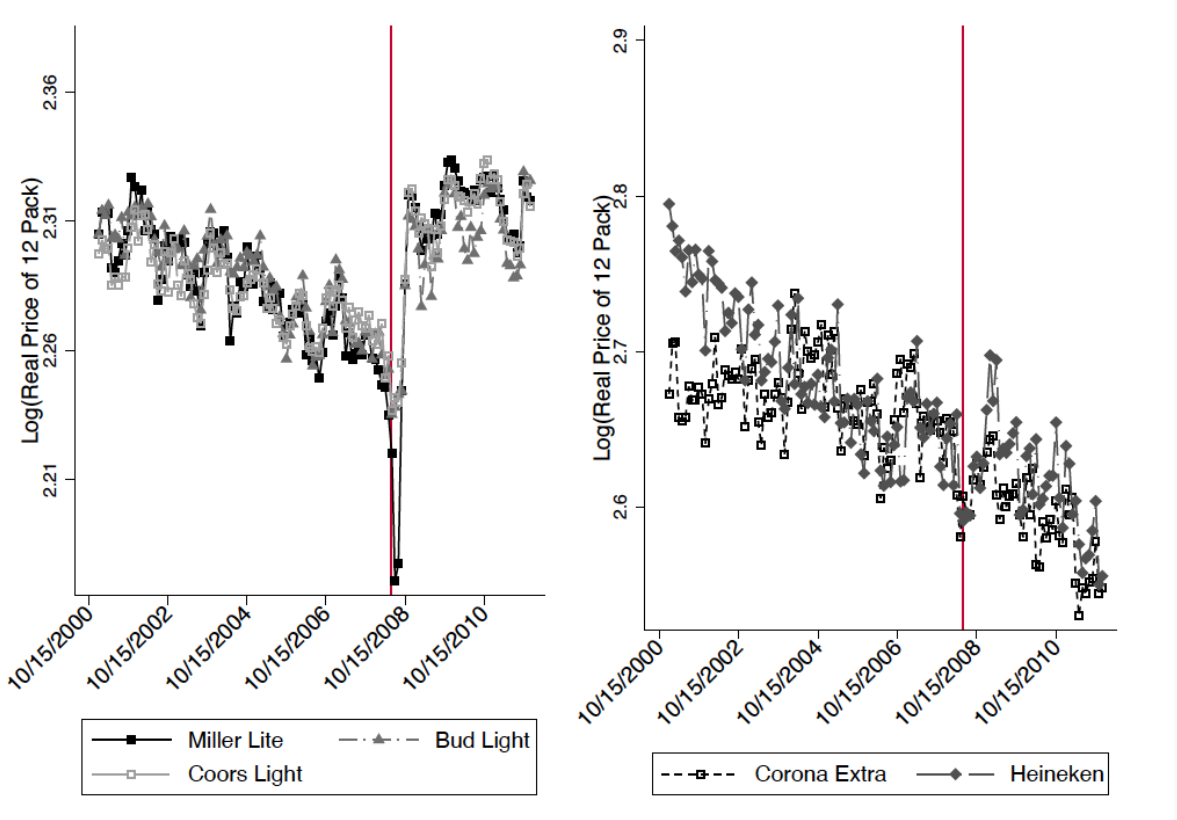
Industry Background

Table I: Revenue Shares and HHI

Year	ABI	MillerCoors	Miller	Coors	Modelo	Heineken	Total	HHI
2001	0.37	.	0.20	0.12	0.08	0.04	0.81	2,043
2003	0.39	.	0.19	0.11	0.08	0.05	0.82	2,092
2005	0.36	.	0.19	0.11	0.09	0.05	0.79	1,907
2007	0.35	.	0.18	0.11	0.10	0.06	0.80	1,853
2009	0.37	0.29	.	.	0.09	0.05	0.80	2,350
2011	0.35	0.28	.	.	0.09	0.07	0.79	2,162

Notes: This table provides revenue shares and the HHI over 2001–2011. Firm-specific revenue shares are provided for ABI, Miller, Coors, Modelo, and Heineken. The total across these firms is also provided. The HHI is scaled from 0 to 10,000. The revenue shares incorporate changes in brand ownership during the sample period, including the merger of Anheuser-Busch (AB) and InBev to form ABI, which closed in November 2008, and Heineken's acquisition of the FEMSA brands in April 2010. All statistics are based on supermarket sales recorded in IRI scanner data.

Price Patterns



- The vertical line is June 2008. Horizontal ticks at October, because brewers reset prices in early autumn.
- Hard to explain by unilateral effects: that would require strong strategic complementarity among domestic brands *and* very weak strategic complementarity between domestic and imported brands — an uncomfortable combination.

Cross-sectional Variation: Is ΔHHI Enough?

- Standard practice assumes price effects are proportional to ΔHHI. This has a basis *if* substitution is proportional to shares: for two merging products $\Delta\text{HHI} = 2s_j s_k$, and the sum of diversion ratios is approximately $s_j + s_k + \Delta\text{HHI}$.
- So run the merger DiD region by region:

$$\log p^R_{jrt} = \alpha_1 \Delta\text{HHI}_r \times 1\{\text{Post}\}_t + \alpha_2 \Delta\text{MILES}_r \times 1\{\text{Post}\}_t + \alpha_3 1\{\text{Post}\}_t + \phi_{jr} + \tau_t + \epsilon_{jr}$$

with ΔMILES_r the reduction in Coors shipping distance (an efficiency).

	Pooled	MillerCoors	ABI	Imports
ΔHHI × Post	0.997 (0.454)	1.172 (0.542)	1.503 (0.531)	−0.005 (0.534)
ΔMILES × Post	−0.042 (0.013)	−0.040 (0.016)	−0.053 (0.013)	−0.028 (0.014)
1{Post}	0.037 (0.012)	0.049 (0.014)	0.040 (0.013)	0.019 (0.014)

- Concentration and distance effects have the right signs but roughly *cancel* on average ($0.997 \times 0.02 \approx 2.0\%$ vs. $-0.042 \times 0.36 \approx -1.5\%$).
- So the **post-merger dummy** carries the action: 4.9% (MillerCoors), 4.0% (ABI). Unilateral effects do not appear to account for the increases — but inferences are weak, since ΔHHI proxies them only under proportional substitution. *Hence the demand model.*

Cross-sectional Variation

Table III: Cross-Sectional Variation in Price Increases

	Pooled	MillerCoors	ABI	Imports
ΔHHI $\times$ $\mathbb{1}$\{Post-Merger\}	0.997	1.172	1.503	-0.005
	(0.454)	(0.542)	(0.531)	(0.534)
ΔMILES $\times$ $\mathbb{1}$\{Post-Merger\}	-0.042	-0.040	-0.053	-0.028
	(0.013)	(0.016)	(0.013)	(0.014)
$\mathbb{1}$\{Post-Merger\}	0.037	0.049	0.040	0.019
	(0.012)	(0.014)	(0.013)	(0.014)
# Observations	167,695	75,315	50,810	41,570

Notes: Estimation is with OLS. The dependent variable is log real retail price. Observations are at the brand–size–region–month–year level. The estimation sample spans 39 regions from 2001 to 2011. All regressions include a linear trend and product (brand $\times$size) fixed effects interacted with region fixed effects. Standard errors are clustered at the region level and shown in parentheses.

- Using regional HHI variation, the authors show *indicative* evidence that unilateral effects are not sufficient to account for the price increase.

The Documentary Record

- The structural test is more persuasive when the institution matches. The DOJ complaint to enjoin ABI's acquisition of Grupo Modelo describes exactly the mechanism:
  - ABI and MillerCoors announce nominal price increases each year in **late summer**, to take effect in **early fall** — matching the October ticks in the price figure.
  - In most geographic areas ABI is the share leader and **announces first**; elsewhere MillerCoors does.
  - Increases are usually **matched by the follower**; if not, they are **rescinded**.
  - ABI's internal "Conduct Plan" is described by DOJ as reading like a how-to manual for price coordination.
- SABMiller and Anheuser-Busch annual reports over 2005–2011 discuss industry pricing conditions in terms consistent with this; ABI's post-merger reports are notably more opaque.
- Methodological point: qualitative evidence does not identify anything on its own, but it disciplines the *modelling choice* — which firms to place in the coalition, and which to treat as a price-taking fringe. Compare Porter (1983), where the institutional record of the Joint Executive Committee plays the same role.

Demand: Random Coefficients Nested Logit

- Regions $r = 1, \ldots, R$, periods $t = 1, \ldots, T$, consumers $i = 1, \ldots, N_{rt}$, inside goods $j = 1, \ldots, J_{rt}$, outside good $j = 0$.

$$u_{ijrt} = x_j \beta_i^* + \alpha_i^* p_{jrt} + \sigma_j^D + \tau_t^D + \xi_{jrt} + \bar{\epsilon}_{ijrt}, \qquad u_{i0rt} = \bar{\epsilon}_{i0rt}$$

- x_j: a constant (inside-good indicator), **calories** (correlated with alcohol content — distinguishes "light" beers), **package size**, **import** dummy. σ_j^D, τ_t^D = product and period fixed effects; ξ_{jrt} = structural error.
- Heterogeneity: $[\alpha_i^*, \beta_i^*]' = [\alpha, \beta]' + \Pi D_i$, with D_i = **demeaned consumer income** (the only source of observed heterogeneity). Since x_j is time-invariant, β is absorbed by the product fixed effects.
- Nested logit error (Berry 1994; Cardell 1997), with group 1 = all inside goods, group 0 = outside good:

$$\bar{\epsilon}_{ijrt} = \zeta_{igrt} + (1 - \rho)\epsilon_{ijrt}, \qquad 0 \leq \rho < 1$$

  Larger $\rho \Rightarrow$ more correlated preferences within the inside nest $\Rightarrow$ *less* substitution to the outside good.
- Market size = 1.5× max observed unit sales per region; the outside good absorbs craft beer, non-supermarket beer, and wine — so their prices are non-strategic.

Demand: Shares and the Nested Logit Special Case

- Decompose indirect utility into mean and consumer-specific parts:

$$\delta_{jrt} = x_j\beta + \alpha p_{jrt} + \sigma_j^D + \tau_t^D + \xi_{jrt}, \qquad \mu_{ijrt} = [p_{jrt}, x_j]' \, \Pi D_i$$

- Market shares integrate over the income distribution:

$$s_{jrt} = \frac{1}{N_{rt}} \sum_{i=1}^{N_{rt}} \frac{\exp\big((\delta_{jrt} + \mu_{ijrt})/(1-\rho)\big)}{\exp\big(I_{i1rt}/(1-\rho)\big)} \cdot \frac{\exp I_{i1rt}}{\exp I_{irt}}$$

  with McFadden inclusive values $I_{i1rt} = (1-\rho)\log\sum_j \exp((\delta_{jrt} + \mu_{ijrt})/(1-\rho))$, $I_{i0rt} = 0$, $I_{irt} = \log(1 + \exp I_{i1rt})$.

- **If $\Pi = 0$** this collapses to the linear nested logit:

$$\log(s_{jrt}) - \log(s_{0rt}) = x_j\beta + \alpha p_{jrt} + \sigma_j^D + \tau_t^D + \rho \log(s_{jrt|g}) + \xi_{jrt}$$

- Why the nest is doing real work here: ρ keeps elasticities from being driven by the (arbitrary) market-size assumption. But nested logit still forces substitution among inside goods to depend only on shares. The income random coefficients break *that* — which matters because domestic-vs-import substitution is exactly what the conduct test hinges on.

Demand: Estimation and Instruments

- BLP nested fixed point / GMM: $\hat{\theta}^D = \arg\min_\theta \ \omega(\theta)' Z A^{-1} Z' \omega(\theta)$. For candidate $(\tilde{\Pi}, \tilde{\rho})$ a contraction delivers δ^* equating predicted and observed shares; $\omega(\tilde{\Pi}, \tilde{\rho})$ is the residual from a 2SLS regression of δ^* on price and fixed effects. α falls out of that 2SLS, so the nonlinear search is only over (Π, ρ). Two-step GMM, cluster-corrected; asymptotics in the number of regions.
- **Three instrument groups** (12 instruments total):
  1. **For price**: (a) distance between brewery and region (miles$\times$ diesel index) — a cost shifter; (b) an indicator for ABI and MillerCoors products post-merger — a change in competitive structure.
  2. **For** ρ: variation in the *conditional* inside share $s_{jrt|g}$. Uses the number of products in the market and total distance summed over all products, interacted with ABI and Miller/Coors indicators.
  3. **For** Π: mean regional income interacted with each product characteristic (constant, calories, package size, import).
- Berry-Haile (2014) point: with random coefficients, price endogeneity is not the only problem — the mapping $\delta^*(x, p, S; \tilde{\Pi}, \tilde{\rho})$ itself depends on the nonlinear parameters, so each needs its own excluded instrument. This is why (2) and (3) are separate groups, not decoration.

Demand Estimates

Table IV: Baseline Demand Estimates

Demand Model:		NL-1	RCNL-1	RCNL-2	RCNL-3	RCNL-4
Data Frequency:		Monthly	Monthly	Quarterly	Monthly	Quarterly
Variable	Parameter	(i)	(ii)	(iii)	(iv)	(v)
Price	α	-0.1312	-0.0887	-0.1087	-0.0798	-0.0944
		(0.0884)	(0.0141)	(0.0163)	(0.0147)	(0.0146)
Nesting Parameter	ρ	0.6299	0.8299	0.7779	0.8079	0.8344
		(0.0941)	(0.0402)	(0.0479)	(0.0602)	(0.0519)
Demographic Interactions						
Income×Price	Π_1		0.0007	0.0009		
			(0.0002)	(0.0003)		
Income×Constant	Π_2		0.0143	0.0125	0.0228	0.0241
			(0.0051)	(0.0055)	(0.0042)	(0.0042)
Income×Calories	Π_3		0.0043	0.0045	0.0038	0.0031
			(0.0016)	(0.0017)	(0.0018)	(0.0015)
Income×Import	Π_4				0.0039	0.0031
					(0.0019)	(0.0016)
Income×Package Size	Π_5				-0.0013	-0.0017
					(0.0007)	(0.006)
Other Statistics						
Median Own Price Elasticity		-3.81	-4.74	-4.33	-4.45	-6.10
Median Market Price Elasticity		-1.10	-0.60	-0.72	-0.60	-0.69
Median Outside Diversion		29.80%	12.96%	16.98%	13.91%	11.82%
J-Statistic			13.94	13.75	13.91	14.15

- RCNL-1: $\hat{\alpha} = -0.0887\,(0.0141)$, $\hat{\rho} = 0.83\,(0.04)$; income $\times$ price, constant, calories all significant.
- Median own-price elasticity -4.74; median *market* elasticity only -0.60. Diversion to the outside good 12–17% (vs. 29.8% under plain nested logit). Consumers substitute among beers, not out of beer.
- Higher-income consumers are less price-sensitive and prefer inside goods, imports, more calories, smaller packages. Hansen J cannot reject validity at 95%.

Why the Substitution Pattern Matters for the Test

- From the full elasticity matrix (RCNL-1, 12-packs):
  - Bud Light buyers divert disproportionately to **Budweiser, Coors Light, Miller Lite** and to larger package sizes. Cross-elasticity to domestic $= 0.349$ vs. imported $= 0.279$.
  - Corona Extra buyers divert disproportionately to **Heineken** and smaller sizes: imported $= 0.301$ vs. domestic $= 0.229$.
  - Diversion to the outside good is tiny (≈ 0.016 for Bud Light).
- **Consequence for unilateral effects**: a Miller/Coors merger raises MillerCoors prices; ABI's best response is to raise its prices too, because domestic brands are strong strategic complements. *Some* ABI price increase is therefore a pure unilateral effect.
- **But**: the estimated complementarity is not strong enough to deliver the *observed* ABI increase, while imports — weak complements — should barely move, and indeed they do not.
- This is the heart of the identification: the demand system quantifies exactly how much of ABI's increase a Bertrand equilibrium can produce. The residual is attributed to conduct. If demand were misspecified toward too little domestic substitution, $\hat{\kappa}$ would be biased *up*.

Supply: The Conduct Parameter κ

- Equilibrium prices in each region-period satisfy

$$p_t = mc_t - \left[\Omega_t(\kappa) \circ \left(\frac{\partial s_t(p_t;\theta^D)}{\partial p_t}\right)^T\right]^{-1} s_t(p_t;\theta^D)$$

  where $\circ$ is element-by-element multiplication. The (j,k) element of Ω_t is 1 if j,k are produced by the same firm; κ if j,k belong to ABI and MillerCoors *post-merger*; 0 otherwise.

- With four products sold by ABI, Miller, Coors, Modelo:

$$\Omega_{t_1^*} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad \Omega_{t_2^*} = \begin{pmatrix} 1 & \kappa & \kappa & 0 \\ \kappa & 1 & 1 & 0 \\ \kappa & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- $\kappa = 0$: post-merger Nash-Bertrand. $\kappa = 1$: joint profit maximization for ABI and MillerCoors. **Key restriction**: Modelo and Heineken price à la Nash — motivated by the documentary evidence, not by the data.

- Note what κ is not: it is not an ownership share and not a merger. It is a wedge in the *off-diagonal* block between two firms that remain separately owned — exactly the object a unilateral-effects simulation sets to zero by assumption.

Deriving the Multi-Product Bertrand FOC

Firm f owns products $\mathcal{J}_f$ and (post-merger) partially internalizes coalition partner g's profits via κ. Firm f's objective in a region-period:

$$\max_{\{p_j\}_{j\in\mathcal{J}_f}} \sum_{k\in\mathcal{J}_f} (p_k - mc_k)\, s_k(p_t) \;+\; \kappa \sum_{k\in\mathcal{J}_g} (p_k - mc_k)\, s_k(p_t)$$

- FOC for product $j \in \mathcal{J}_f$: $\; s_j(p_t) + \sum_{k\in\mathcal{J}_f} (p_k - mc_k)\frac{\partial s_k}{\partial p_j} + \kappa \sum_{k\in\mathcal{J}_g} (p_k - mc_k)\frac{\partial s_k}{\partial p_j} = 0$
- Collect all J such FOCs into a single system. The coefficient on $(p_k - mc_k)$ in product j's FOC is exactly $\Omega_{jk}(\kappa)$: one if j, k share a firm, κ if j, k are ABI/MillerCoors post-merger, zero otherwise.
- In matrix form, stacking over j:

$$s_t(p_t) + \left[\Omega_t(\kappa) \circ \left(\frac{\partial s_t(p_t)}{\partial p_t}\right)^T\right](p_t - mc_t) = 0$$

- Solving for p_t: $\; p_t = mc_t - \left[\Omega_t(\kappa) \circ (\partial s_t/\partial p_t)^T\right]^{-1} s_t(p_t)$.
- Compare with GMY: identical machinery, but there Ω was known and mc was the unknown. Here we ask the data to deliver *both* — which is why we need an extra exclusion restriction.

Supply: Marginal Cost and Estimation

- Parameterize marginal cost of product j in region r, period t:

$$mc_{jrt} = w_{jrt}\gamma + \sigma_j^S + \tau_t^S + \mu_r^S + \eta_{jrt}$$

where w_{jrt} contains (i) distance (miles$\times$ diesel index) and (ii) an indicator for MillerCoors products post-merger. So the merger can lower costs through **distribution rationalization** and through **residual synergies** unrelated to distance.

- Taking $\hat{\theta}^D$ as given, the supply-side structural error is

$$\eta_{rt}^*(\tilde{\theta}^S; \hat{\theta}^D) = p_{rt} - w_{jrt}\tilde{\gamma} - \sigma_j^S - \tau_t^S - \mu_r^S + \left[\Omega_t(\tilde{\kappa}) \circ \left(\frac{\partial s_t(p_{rt}; \hat{\theta}^D)}{\partial p_{rt}}\right)^T\right]^{-1} s_t(p_{rt}; \hat{\theta}^D)$$

and $\hat{\theta}^S = \arg\min_\theta \ \eta^*(\theta; \hat{\theta}^D)' zz' \eta^*(\theta; \hat{\theta}^D)$.

- Fixed effects and γ are concentrated out by OLS, so the nonlinear search is one-dimensional in κ. Standard errors clustered by region, with a Wooldridge (2010) correction for the first-stage demand estimates.

- **Maintained assumption**: constant marginal cost. If ABI's *mc* sloped *up*, then as consumers shift to ABI it would raise prices for cost reasons too, and $\hat{\kappa}$ would be biased **upward**. The authors argue ABI faced no capacity constraints in the sample.

Discussion of Identification

- The markup term in η^* is **endogenous**: unobserved costs enter implicitly through price. Instrument with an indicator for ABI and MillerCoors products in post-merger periods.
- **Validity condition**: given product, region, and time fixed effects, changes in ABI's unobserved costs before vs. after the merger are not systematically different from changes for Modelo and Heineken. The separate MillerCoors post-merger cost dummy isolates the comparison to **ABI vs. imports**.
- **What variation moves what.** $\hat{\kappa} > 0$ iff ABI's post-merger prices exceed what unilateral effects can explain. But a positive κ raises *both* firms' prices — so $\hat{\gamma}_1 < 0$ iff MillerCoors prices are *lower* than unilateral effects plus κ would imply. The two parameters are pinned down by the **relative** movement of ABI and MillerCoors prices.

Identification: A Calibrated Illustration

- Logit demand, three symmetric firms, $p = 1.00$, $mc = 0.70$, $s_j = 0.20$, $s_0 = 0.40$. Let firms 1 and 2 merge:

Scenario	Post-merger prices
Nash-Bertrand ($\kappa = 0$)	(1.06, 1.06, 1.01)
$\kappa = 0.50$	(1.10, 1.10, 1.07)
$\kappa = 0.50$ and $mc \to 0.50$ for merging firms	(1.09, 1.09, 1.08)

  Note the **non-merging** firm's price is the informative one: $1.01 \to 1.07 \to 1.08$.

- This is Bresnahan's rotation argument in disguise: the merger shifts rivals' marginal revenue, and the focal firm's response separates conduct from cost. Compare the Firm Conduct lecture, and Berry-Haile (2014) on differentiated products.

Supply Estimates

Table VI: Baseline Supply Estimates

Demand Model: Data Frequency: Variable	Parameter	NL-1 Monthly (i)	RCNL-1 Monthly (ii)	RCNL-2 Quarterly (iii)	RCNL-3 Monthly (iv)	RCNL-4 Quarterly (v)
Post-Merger Internalization of Coalition Pricing Externalities	κ	0.374 (0.034)	0.264 (0.073)	0.249 (0.087)	0.286 (0.042)	0.342 (0.054)
Marginal Cost Parameters						
MillerCoors×PostMerger	γ_1	-0.608 (0.039)	-0.654 (0.050)	-0.649 (0.060)	-0.722 (0.042)	-0.526 (0.040)
Distance	γ_2	0.142 (0.046)	0.168 (0.059)	0.163 (0.059)	0.169 (0.060)	0.148 (0.049)

Notes: This table shows the baseline supply results. We use the method of moments for estimation. There are 94,656 observations at the brand–size–region–month–year level in columns (i), (ii), and (iv) and 31,784 observations at the brand–size–region–year–quarter level in columns (iii) and (v). The samples exclude the months/quarters between June 2008 and May 2009. All regressions include product (brand×size), period (month or quarter), and region fixed effects. Standard errors clustered by region and shown in parentheses.

- $\hat{\kappa}$ is positive and significant in *every* specification: 0.249–0.342 across the RCNL models (0.264 with standard error 0.073 in the baseline RCNL-1). Post-merger Nash-Bertrand is easily rejected.
- MillerCoors post-merger cost shift $\hat{\gamma}_1 \in [-0.72, -0.53]$, i.e. efficiencies of about \$0.66–\$0.70 per unit — roughly a 14% reduction in the marginal cost of Coors Light.
- Distance $\hat{\gamma}_2 \approx 0.15$–$0.17$: shipping-related marginal cost is 2–3% of retail price.

Implied Brewer Markups

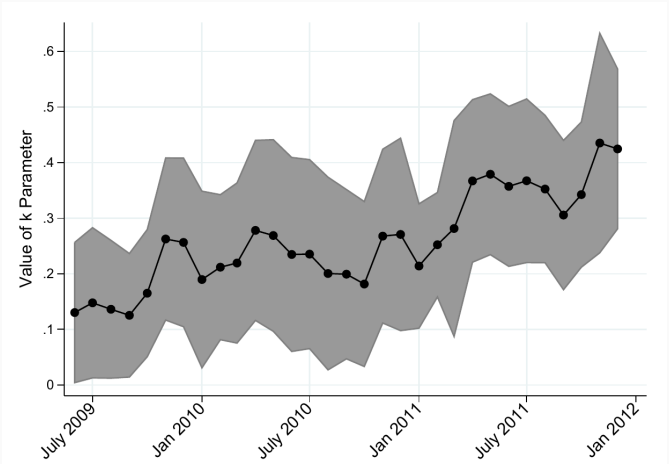
	6 packs		12 packs		24 packs	
Brand	Pre	Post	Pre	Post	Pre	Post
Bud Light	3.63	4.34	3.52	4.24	3.43	4.13
Coors Light	2.47	4.21	2.36	4.14	2.28	4.04
Miller Lite	2.89	4.25	2.78	4.18	2.69	4.07
Corona Extra	3.30	3.18	3.04	2.91	3.04	3.03
Heineken	3.20	3.14	2.98	2.92	3.22	3.33

Average markups ($/equivalent unit), RCNL-1. Full table in the paper covers 13 brands.

- Average markup across all 94,656 observations is $3.60, about 34% of retail price.
- ABI 12-packs: markups rise about $0.70. Miller: +$1.40. Coors: +$1.80. **Imports: essentially unchanged.** The larger Miller/Coors increases combine *higher prices* with *lower marginal costs* — efficiencies show up here, not in ABI.
- Connect back to Part I: a production-approach estimator would see this markup rise and, with no demand system, attribute it to technology or the input share. Softer competition is invisible to it.

Was It the Merger, or the Recession?

- Identification threat: the merger coincides with the Great Recession, and Rotemberg-Saloner (1986) predicts **countercyclical** collusion. ABI's 2009 report calls it "the most difficult" environment in many years.
- Three pieces of evidence against the recession story:
  1. ABI/MillerCoors prices keep rising relative to imports over 2009–2011, during *recovery*, and are *positively* associated with household earnings.
  2. Estimating a separate κ for each post-merger month gives estimates that **increase** over time — the opposite of the Rotemberg-Saloner prediction.
  3. Letting κ vary with local income via a logistic link gives $\hat{a} = 0.456\,(0.088)$ on the post-merger dummy, $\hat{b} = 0.033\,(0.039)$ on mean income: no countercyclicality in the cross-section either.



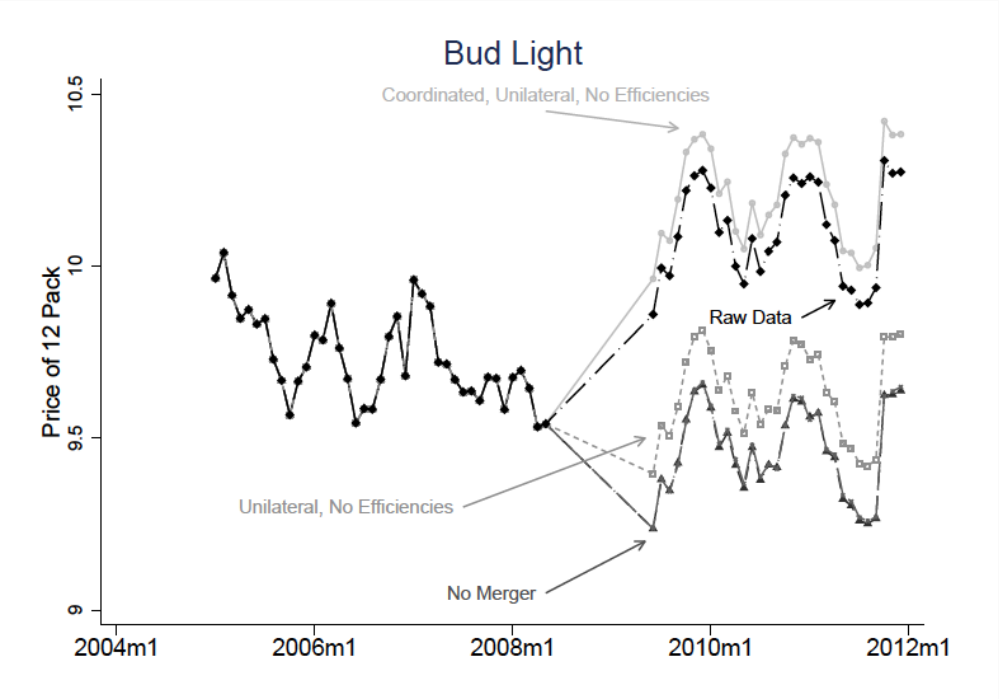
Counterfactual Design

Recompute equilibrium under the RCNL-1 estimates, switching three mechanisms on and off:

Scenario	Coordinated	Unilateral	Efficiencies
(i) Raw data / baseline	yes	yes	yes
(ii) Merger, no efficiencies	yes	yes	no
(iii) Merger, no coordination	no	yes	yes
(iv) Merger, neither	no	yes	no
(v) No merger	no	no	no

- **Unilateral** = Miller and Coors internalizing competition with each other (Ω block goes to ones).
- **Coordinated** = the non-zero post-merger κ between ABI and MillerCoors.
- **Efficiencies** = the estimated $\hat{\gamma}_1$ cost reduction plus shorter shipping distances.
- Why this decomposition is the policy-relevant one: (iii) is very plausibly the scenario the DOJ had in mind when it cleared the joint venture.

Counterfactual Price Trajectory



- For Miller Lite, unilateral effects and efficiencies roughly **cancel**: "No Merger" and "Unilateral, Efficiencies" nearly coincide. Both are individually large.
- Coordinated effects are strong enough that prices rise *despite* the large cost reductions — about +50 cents per 12-pack relative to no merger.
- For Bud Light the post-merger increase is **almost entirely** coordinated effects, since ABI gets no efficiencies at all.

Counterfactual Welfare (2011)

Coordinated:	yes	yes	no	no	no
Unilateral:	yes	yes	yes	yes	no
Efficiencies:	yes	no	yes	no	no
	(i)	(ii)	(iii)	(iv)	(v) No merger
Retail prices, 12 packs					
ABI	10.03	10.14	9.38	9.55	9.43
Coors	10.18	10.85	9.56	10.22	9.26
Welfare, % change vs. "No merger"					
Producer surplus	+22.1	+19.1	+10.3	+8.2	—
Consumer surplus	−3.7	−5.3	−0.2	−2.1	—
Total surplus	+1.3	−0.6	+1.8	−0.1	—

- Prices are 6–8% higher, markups 17–18% higher, than post-merger Nash-Bertrand ((i) vs. (iii)). Producer surplus +22.1%, **over half of it from coordination**.
- Consumer surplus falls 3.7% — but only 0.2% without coordinated effects (iii). The consumer harm is essentially all coordinated effect.
- Total surplus still rises 1.3%: clearance was defensible under a *total* welfare standard, not under the consumer standard US agencies actually apply.

How Much Should We Believe κ?

- **What κ is**: a reduced-form index of internalization, interpretable as a conduct parameter under conditions in Black-Crawford-Lu-White (2004) and Sullivan (2016).
- **Corts (1999) critique**: if true conduct varies *within* a regime, the conduct-parameter interpretation is problematic — κ is then some average whose mapping to any underlying game is unclear. This is why the paper frames the exercise as a **test** of post-merger Nash-Bertrand rather than a structural estimate of collusion.
- **What is genuinely assumed, not estimated**:
  - The coalition membership (ABI + MillerCoors) and that Modelo/Heineken are Nash — from the documentary record.
  - Constant marginal cost (biases $\hat{\kappa}$ up if false).
  - That the demand system is right; too little estimated domestic substitution also biases $\hat{\kappa}$ up.
  - A constant-markup brewer-retailer structure; marginal cost *levels* are sensitive to retail market power, brewer *markups* are not.
- **What the paper does not do**: explain *why* coordination became feasible. That is the open question, and what a fully dynamic repeated-game treatment would have to supply.

Case Study 2: What to Take Away

- **Conduct is estimable**, at least as a test, when a merger supplies exogenous variation in the ownership matrix. This is the Bresnahan/Berry-Haile identification logic applied to a differentiated-product market with a regime shift.
- **Unilateral-effects analysis can be badly incomplete.** Here it would have predicted roughly no price effect once efficiencies are counted — and prices rose 6–8%.
- **Efficiencies do not settle the question.** The MillerCoors cost savings were real and large, and consumers still lost, because the merger changed how the *remaining rivals* competed.
- **The right welfare standard matters.** Total surplus rises; consumer surplus falls. The two standards give opposite answers on the same estimates.
- And back to the aggregate debate: here is a well-identified, economically large increase in markups in a major consumer-goods industry — driven by *conduct*, invisible to a production-function estimator, and averaged into near-nothing in an economy-wide series. "No aggregate trend" is not "no market power."

Three Lenses on the Same Question

	Production approach DEU, Raval, Demirer	**Structural IO, conduct assumed** GMY (autos)	**Structural IO, conduct estimated** MW (beer)
mc from	technology + cost min.	demand + Nash-Bertrand	demand + $\Omega(\kappa)$
Needs demand?	no	yes	yes
Conduct	not modelled	*assumed* static B-N	*tested and rejected*
Identified by	input shares	RXR cost shifter	merger shift in Ω
Quality change	invisible	measured (τ_t)	not at issue (11 yrs)
Welfare	not available	*CS*, *PS*, *DWL* levels	ΔCS, ΔPS vs. co factual
Scope	whole economy	one industry, 39 yrs	one industry, one m

- The trade-off is **coverage vs. credibility**, and better econometrics alone will not resolve it. The complementarity: industry studies are how we *validate* the aggregate estimator. De Loecker and Scott (2022) find the two agree in beer; GMY find they disagree in autos, and can say why.

Summary

- Rising aggregate markups are an important hypothesis, so far with not-so-obvious empirical support: the *level* is close to unidentified from revenue data (Bond et al.); the *trend* is fragile to the flexible-input definition (Traina) and to input-augmenting technology (Raval, Demirer); and the welfare-relevant object is markup *dispersion*, not the level (EMX).
- IO economists therefore look for "smoking-gun evidence" in detailed industry studies:
  - Grieco, Murry, and Yurukoglu (2024), automobiles: markups **fall** by roughly half; welfare gains come from quality and production technology.
  - Miller and Weinberg (2017), beer: markups rose 17–18% with the MillerCoors joint venture — through *coordination*, a mechanism no production-function estimator can see.
  - Dopper, MacKay, Miller, and Stiebale (2024), consumer products: some evidence of rising markups 2006–2019, mostly from changing consumer preferences.
- Bottom line: left to be convinced that aggregate markups changed dramatically (and drove much of the labor share decline). But "no aggregate trend" is not "no market power."

References i

The aggregate markup debate

[1] Bond, S., A. Hashemi, G. Kaplan, and P. Zoch (2021). "Some Unpleasant Markup Arithmetic: Production Function Elasticities and Their Estimation from Production Data." *Journal of Monetary Economics* 121, 1–14.

[2] De Loecker, J., J. Eeckhout, and G. Unger (2020). "The Rise of Market Power and the Macroeconomic Implications." *Quarterly Journal of Economics* 135(2), 561–644.

[3] De Loecker, J., and F. Warzynski (2012). "Markups and Firm-Level Export Status." *American Economic Review* 102(6), 2437–2471.

[4] Demirer, M. (2025). "Production Function Estimation with Factor-Augmenting Technology: An Application to Markups." Working paper.

References ii

[5] Edmond, C., V. Midrigan, and D. Y. Xu (2023). "How Costly Are Markups?" *Journal of Political Economy* 131(7), 1619–1675.

[6] Hall, R. E. (1988). "The Relation between Price and Marginal Cost in U.S. Industry." *Journal of Political Economy* 96(5), 921–947.

[7] Klette, T. J., and Z. Griliches (1996). "The Inconsistency of Common Scale Estimators When Output Prices Are Unobserved and Endogenous." *Journal of Applied Econometrics* 11(4), 343–361.

[8] Raval, D. (2023). "Testing the Production Approach to Markup Estimation." *Review of Economic Studies* 90(5), 2592–2611.

[9] Traina, J. (2018). "Is Aggregate Market Power Increasing? Production Trends Using Financial Statements." Stigler Center New Working Paper Series No. 17.

References iii

Industry case studies (this lecture)

[1] Grieco, P. L. E., C. Murry, and A. Yurukoglu (2024). "The Evolution of Market Power in the US Automobile Industry." *Quarterly Journal of Economics* 139(2), 1201–1253. `[IO Literature/Monopoly Power/]`

[2] Miller, N. H., and M. C. Weinberg (2017). "Understanding the Price Effects of the MillerCoors Joint Venture." *Econometrica* 85(6), 1763–1791. `[Monopoly Power/mwbeer.pdf]`

[3] Dopper, H., A. MacKay, N. H. Miller, and J. Stiebale (2024). "Rising Markups and the Role of Consumer Preferences." Working paper.

Background referenced in passing

References iv

[1] Ackerberg, D. A., K. Caves, and G. Frazer (2015). "Identification Properties of Recent Production Function Estimators." *Econometrica* 83(6), 2411–2451.

[2] Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020). "The Fall of the Labor Share and the Rise of Superstar Firms." *Quarterly Journal of Economics* 135(2), 645–709.

[3] Berry, S., J. Levinsohn, and A. Pakes (1995). "Automobile Prices in Market Equilibrium." *Econometrica* 63(4), 841–890.

[4] Berry, S., J. Levinsohn, and A. Pakes (2004). "Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market." *Journal of Political Economy* 112(1), 68–105.

References v

[5] Corts, K. S. (1999). "Conduct Parameters and the Measurement of Market Power." *Journal of Econometrics* 88(2), 227–250.

[6] De Loecker, J., and P. T. Scott (2022). "Markup Estimation Using Production and Demand Data: An Application to the US Brewing Industry." Working paper.

[7] Pakes, A., S. Berry, and J. Levinsohn (1993). "Applications and Limitations of Some Recent Advances in Empirical Industrial Organization: Price Indexes and the Analysis of Environmental Change." *American Economic Review P&P* 83(2), 240–246.

[8] Porter, R. H. (1983). "A Study of Cartel Stability: The Joint Executive Committee, 1880–1886." *Bell Journal of Economics* 14(2), 301–314.

References vi

[9] Rotemberg, J. J., and G. Saloner (1986). "A Supergame-Theoretic Model of Price Wars during Booms." *American Economic Review* 76(3), 390–407.